

Demazure's structure theorem for the automorphism group of a toric variety: old and new

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Abstract

This article provides an accessible and simplified proof of Demazure's 1970 structure theorem for the automorphism group of a compact smooth toric variety, previously considered quite inaccessible. Moreover, we simplify Cox's extension of Demazure's theorem to orbifolds, obtained over the years 1995–2014, allowing us to generalize it to any compact (arbitrarily singular) toric variety and toric log pairs. Finally, we further generalize the theorem to non-compact toric varieties obtained as the complement of toric subvariety of a codimension at least 2, resolving a problem originating in Oda's classic 1988 book.

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In this paper, all the varieties are defined over $\mathbb{C}$.

1 A forgotten theorem

Symmetry in geometry is often measured and quantified via automorphism groups of geometric structures: the larger and richer in structure the more aesthetically pleasing. The automorphism group of a complex (algebraic or analytic) variety X , denoted by $\text{Aut } X$ (Definition 2.18), consists of all automorphisms of X (biholomorphisms if X is a complex manifold). It is a (complex) Lie group when X is proper (compact in complex analytic topology) [4, p. 87]. If X further possesses a *toric* structure (Definition 2.1) it is natural to expect that $\text{Aut } X$ enjoys even more structure.

That turns out to be true and is a fundamental fact in the theory of toric varieties. But, it is far from simple to prove.

A caveat is that maps in $\text{Aut } X$ *need not preserve the toric structure*. Thus, the analysis of $\text{Aut } X$ is not as simple as one might initially expect.

The celebrated Demazure Structure Theorem for $\text{Aut } X$ when X is a smooth proper toric scheme over $\mathbb{Z}$, originally formulated in 1970 [12], gives a rather granular description of $\text{Aut } X$ as a linear algebraic group, with the different connected components of $\text{Aut } X$ corresponding to a quotient group of $\text{Aut } \Delta$, where Δ is the associated fan (Definition 2.3 and Definition 2.20).

Even now, more than half a century after Demazure’s article, the proof of the Structure Theorem is (as far as we are aware) not featured in any book on toric varieties [21, 26, 28, 15, 14, 13, 8, 11]. Perhaps even more astonishing, it is only *stated* in one of these books, Oda’s 1988 classic (and Oda has some interesting remarks on it... see below). This is even more remarkable, since Demazure’s article was the first on toric varieties, and essentially initiated the field, and this is one of the basic, fundamental, theorems in the field.

Another curious point is that Demazure works in Bourbaki-style generality with schemes throughout his foundational lengthy monograph, but when it comes to analyzing $\text{Aut } X$ he makes the uncharacteristic assumption that X is *smooth*. Cox, between the years 1995–2014, developed a beautiful alternative approach to the theorem that extended it to *simplicial toric variety* (equivalently, *orbifolds*), but still quite far from satisfactory generality.

1.1 Some history

Given these remarks, it seems only natural to delve a bit deeper into the history of this problem. We hope this is instructive to our readers, as it also serves to motivate the present article.

In 1970, Demazure presented the eponymous structure theorem for the automorphism group of a *smooth* compact toric variety: $\text{Aut}_0 X$ (the identity component of $\text{Aut } X$ (Definition 2.18)) is generated by the complex torus $(\mathbb{C}^*)^n$ acting on X as a maximal torus and unipotent subgroups corresponding to what Demazure coined as the *roots* of X , denoted by $R(\Delta)$ (Definition 2.15) [12].

However, for many years, Demazure’s structure theorem remained rather impenetrable, even to many experts. To wit, Oda, in his classical 1988 treatise on toric varieties, omits the proof, commenting [28, p. 140],

“Demazure states his results in this final form in [12, Proposition 11, p. 581] although the proof, which is valid for (not necessarily compact) non-singular toric varieties, is spread over the entire paper. We omit the proof here, since it requires so much space and a basic knowledge of linear algebraic groups.”

(In fact, Demazure’s paper is over 80 pages long. We revisit this quote in a moment.)

In 1995, Cox revisited the problem and provided a new point of view on Demazure’s theorem via the *homogeneous coordinate ring* (Definition 3.4). The advantage of this ring is that it is intrinsic to X and not associated with a particular projective embedding of X [9, §4]. This also showed that Demazure’s theorem extends verbatim to the orbifold case, also referred to as the *simplicial* case by algebraic geometers (Definition 2.7).

Cox’s treatment, while more concise and accessible than Demazure’s, is still technical. Additionally, it originally contained an oversight (See Remark 3.16), which was fixed by Cox in his erratum in 2014 [10], once again emphasizing the subtle and difficult nature of the theorem.

1.2 Why revisit?

In revisiting Demazure–Cox’s theorem we had two main motivations:

- Simply put, this is a beautiful theorem, and when we asked around experts, it seemed that none (except D. Cox) were familiar with the proof—it has become a mostly forgotten proof. This is partly explained by the formal style of [12] (Demazure was part of the Bourbaki collective), the fact that toric varieties were not yet quite even formalized at the time of Demazure’s article (this is 20 years before Oda’s and Fulton’s monographs [28, 15], in fact, Demazure’s results on automorphism groups “seem to have motivated him to initiate the theory of toric variety” as Oda writes [28, p. 135], cf. [11, Appendix A]) but also by the long-winded nature of the argument in [12], that intertwines the study of automorphism groups of toric varieties with the analysis of Cremona transformations (see [28, §3.4], [11, Appendix A] for the connection). Aside from making the proof rather inaccessible, it even sows some confusion regarding the generality of some of the arguments (e.g., see §1.4). The purpose of §§3–4 is to both resurrect this theorem, simplify its proof and make it as geometric and intuitive as possible, along with examples at various stages of the proof. (See Remark 3.16 for some simplifications in the proof compared to the original treatments.) Overall, we present an almost self-contained proof of the Demazure–Cox structure theorem for the automorphism group of a compact toric variety that is accessible to a graduate student in complex/algebraic geometry.

- The special case of Demazure’s theorem when $\mathrm{Aut}_0 X$ is reductive is often used by complex/algebraic geometers in the context of K-stability and is needed in a companion paper [20]. It is stated in Corollary 3.15. We felt it would not be right to simply invoke a theorem whose proof neither we nor anybody we asked in the community knew.

As an added bonus, we also:

- extend the theorem to *all* compact toric varieties, i.e., allow *arbitrary singularities*, and in particular remove the orbifold assumption imposed by Cox.
- extend the theorem to toric *log pairs* (Definition 5.1)
- extend the theorem to certain *incomplete* cases (see Theorem 1.3), resolving a problem originating in Oda’s 1988 book.

1.3 The statements

Overall, the statement of Demazure's theorem (with Cox's improvement) is that, for a simplicial (Definition 2.9) toric variety X defined by a fan Δ (Definition 2.4), $\text{Aut } X$ (Definition 2.18) is generated by the maximal torus, certain *root automorphisms* associated with the lattice points in the relative interior of the codimension 1 faces of the polytope in Definition 2.13, and automorphisms induced by symmetries of the fan Δ . We generalize Cox's methods to remove the simplicial/orbifold condition (see §1.5). Thus, the Structure Theorem generalizes to all compact toric varieties. As far as we are aware, this is the first time this theorem is stated and proved in this generality.

Theorem 1.1. (The Structure Theorem for $\text{Aut } X$) *Let $X = X_\Delta$ (Definition 2.4) be a complete toric variety (Definitions 2.1) over $\mathbb{C}$ defined by a fan Δ (Definition 2.3 and Definition 2.4). Then the automorphism group $\text{Aut } X$ (Definition 2.18) is a linear algebraic group (Definition 2.25), and the identity component $\text{Aut}_0 X$ of $\text{Aut } X$ satisfies:*

- (i) $(\mathbb{C}^*)^n$ is a maximal algebraic torus in $\text{Aut}_0 X$. $\text{Aut}_0 X$ is the semi-direct product of its unipotent radical with a reductive group, which is generated by the maximal algebraic torus and the one-parameter groups corresponding to the semisimple roots $R_s(\Delta)$ (Definition 2.15).
- (ii) The unipotent radical of $\text{Aut}_0 X$ is generated by the one-parameter groups $\phi^{m,\lambda}$ for $m \in R_u(\Delta)$ (see Definition 2.15 and Definition 4.17).
- (iii) $\text{Aut } \Delta$ (Definition 2.20) can be naturally embedded in $\text{Aut } X$ such that $\text{Aut } X$ is the product $\text{Aut } \Delta \text{Aut}_0 X$, and $\text{Aut } \Delta \cap \text{Aut}_0 X \cong \text{Aut}_0 \Delta$ (Definition 2.24).

An even more precise version is stated in Theorem 3.14. For an even more general result for toric log pairs, see Theorem 1.2.

In fact, our proof generalizes to also compute the automorphism group of a toric *log pair* (Definition 5.1). We are grateful to R. Berman for asking us whether such a statement holds, motivating us to include this extension. For the proof of the following theorem, see Subsection 5.1.

Theorem 1.2. (The automorphism groups for toric log pairs) *Let $X = X_\Delta$ be a complete toric variety of dimension n (Definition 2.10) defined by a fan Δ (Definition 2.4), P be the associated polytope (Definition 2.13), and $\mathcal{D} = \sum_{v \in \Delta_1} a_v D_v$ (Definition 2.12). Then $\text{Aut}(X, \mathcal{D})$ (Definition 5.1) is a linear algebraic group that satisfies:*

- (i) $\text{Aut}_0(X, \mathcal{D})$ (Definition 5.1) is a linear algebraic group (Definition 2.25) with a maximal torus $T'_X \cong (\mathbb{C}^*)^n$ (Definition 3.6).
- (ii) The unipotent radical (Definition 2.25) of $\text{Aut}_0(X, \mathcal{D})$ is generated by the one-parameter groups $\phi^{m,\lambda}$ (Definition 4.17) associated with the roots $m \in R_u(X, \mathcal{D})$ (Definition 5.2). $\text{Aut}_0(X, \mathcal{D})$ is the semi-direct product of its unipotent radical with a group G''_s/G , where $G''_s \cong \prod_{\alpha \in C} \text{GL}(|\Delta'_\alpha|, \mathbb{C}) \times (\mathbb{C}^*)^{|\Delta_\alpha \setminus \Delta'_\alpha|}$ (Definition 5.4) and G (given in Definition 3.6) is naturally embedded in G''_s .
- (iii) Further assume $a_v \neq a_{v'}$ when $v \neq v'$ and $a_v a_{v'} \neq 0$. Then $\text{Aut}(\Delta, \mathcal{D})$ (Definition 2.20) can be naturally embedded in $\text{Aut}(X, \mathcal{D})$ such that

$$\text{Aut}(X, \mathcal{D}) / \text{Aut}_0(X, \mathcal{D}) \cong \text{Aut}(\Delta, \mathcal{D}) / (\text{Aut}(\Delta, \mathcal{D}) \cap \text{Aut}_0(X, \mathcal{D})).$$

Finally, we come back to the discussion about Oda's problem mentioned in Section 1.4. Our proof of Theorem 1.1 readily generalize to resolve Oda's problem. Thus, we obtain the structure theorem to certain *incomplete* toric varieties, namely, the complement of a toric invariant closed subset of codimension at least 2 in a complete toric variety. For a sketch of proof, see Subsection 5.2.

Theorem 1.3. *Let $X' := X_{\Delta'}$ be an open toric-invariant subvariety of a toric variety $X := X_{\Delta}$ such that $\text{codim} X \setminus X' \geq 2$, then $\text{Aut} X'$ (Definition 4.6) is a linear algebraic group. The unipotent radical (Definition 2.25) of the identity component $\text{Aut}_0 X'$ is generated by the one-parameter groups $\phi^{m,\lambda}$ (Definition 4.17) associated with the roots $m \in R_u(\Delta')$ (Definition 5.8). The reductive part of $\text{Aut}_0 X'$ is generated by the maximal torus and the one-parameter groups $\phi^{m,\lambda}$ associated with the roots $m \in R_s(\Delta')$ (5.8).*

1.4 Oda’s mysterious remark: compactness

Demazure works with proper schemes [12, Proposition 4, p. 561] when he stated the structure theorem [12, Proposition 11, p. 581], that in our language translates to X being compact (Theorem 2.9) and he also imposes smoothness [12, Proposition 1, p. 558].

It seems from Oda’s remarks, reproduced above, that (a) for many years Demazure’s article was hard to read even for experts, (b) he believed Demazure’s theorem should hold also for non-compact X . There seems to be consensus on (a), especially given the literature mentioned earlier. But what to make of (b)?

Oda’s mentioning of the non-compact case is a bit perplexing, and seems to suggest one of the following¹: either (i) Demazure’s proof should hold also for non-compact smooth toric varieties, or (ii) Demazure’s proof contains gaps (since, as the simplest example shows (in a moment), the theorem fails in the non-compact case!), or (iii) Oda was not aware of that example, or (iv) Oda had in mind that parts of Demazure’s argument are carried out in a broader setting than the final compact statement of the theorem, without intending to assert that the theorem itself extends to arbitrary non-compact varieties, or (v) Demazure’s proof works for some varieties “very close” to compact, e.g., $\mathbb{P}^2$ minus some toric points.

Based on the simplest possible example $X = \mathbb{C}^2$ (Example 1.4), option (i) is off the table, which leads us to believe (iii) is rather unlikely; discussions with experts suggest that (ii) is not probable either.

Here is one way to interpret (iv)–(v). Noting that a toric variety defined by a fan (Definition 2.4) is always normal (all the local rings $\mathcal{O}_x$ are integrally closed) by [11, Theorem 3.1.5], if X' is obtained by removing some closed subvariety of a compact toric variety X with $\text{codim} X \setminus X' \geq 2$, then the automorphism group of X' is a subgroup of $\text{Aut } X$ by Hartogs extension theorem (see the proof of Theorem 1.3).

If X' is toric-invariant, then it is a toric variety defined by a fan Δ' (not complete) with the same 1-dimensional cones of Δ . Noting that Demazure has defined the roots for incomplete fans (see Definition 5.8). In this case, the definition reads as: if $m \in R(\Delta)$ (Definition 2.15) satisfies $\langle m, v \rangle = -1$ for $v \in \Delta_1$ (Definition 2.12), then $m \in R(\Delta')$ if and only if for any $\sigma \in \Delta'$ lying in $m^\perp$, the cone spanned by σ and $\mathbb{R}_+ v$ is also contained in Δ' . This condition is fulfilled automatically when the fan Δ is complete.

There are examples suggesting that in this case $\text{Aut } X'$ should be generated by the maximal torus and the one-parameter subgroups corresponding to the roots in $R(\Delta')$. For example, if X is $\mathbb{P}^2$ minus one/two/three (toric fixed) points, then $\text{Aut } X$ is the subgroup of $\mathbb{P}^2$ that fixes or permutes those points, and is isomorphic to the automorphism group of the one/two/three-point blow-up of $\mathbb{P}^2$ (for more details, see Example 3.18, 3.20, and 3.21).

In fact, this does yield a generalization of Demazure’s theorem to this non-compact setting where a toric codimension 2 subvariety is removed. We state this in Theorem 1.3.

¹To be prudent, we obtained an independent translation of this passage from Oda’s original Japanese text [27], as follows: “This final form can be found in [12, Proposition 11, p. 581], and the proof is developed throughout the paper, including a discussion of a more general setting that is not necessarily restricted to the compact case.” (We are grateful to S. Saito for this.) As the reader can see the two translations are consistent on point (b).

As pointed out to us by Cox, it seems that indeed Theorem 1.3 is a part of what Oda (probably) had in mind in his perplexing comment, since two pages later in his book he notes that Demazure proves that any connected algebraic subgroup of a Cremona group is conjugate to *an* automorphism group of a (not necessarily compact) non-singular toric variety [28, p. 142]. However, Oda does not make it clear if this *subgroup* is *the* whole automorphism group of any particular toric variety. In fact, in the case of Theorem 1.3, this group coincides with the identity component of the automorphism group, but in more general cases, we can only prove that the automorphism group has a subgroup that is generated by the maximal torus and the one-parameter subgroups corresponding to the roots (in the sense of Definition 5.8).

Finally, for added context, it is worth stating here the fact that Demazure’s article was the first on toric varieties and only later [21, 26, 28, 15] the proper notational subtleties involving the fan we used today (which is equivalent to the “éventail” Demazure defined, see Remark 2.6) and “behavior at infinity” were fully developed.

Example 1.4. A simple example that shows Demazure’s theorem may *not be valid* for general non-compact X : the group $\text{Aut} X$ is already not a linear algebraic group for $X = \mathbb{C}^2$! Indeed, consider the subgroup U of $\text{Aut } \mathbb{C}^2$ consisting of the maps $(z, w) \mapsto (z, w + f(z))$ where $f(z) \in \mathbb{C}[z]$. The inverse of such a map is $(z, w) \mapsto (z, w - f(z))$, so it is invertible. Then, U is isomorphic to $\mathbb{C}[z]$ (as an additive group), so U is *infinite-dimensional* over $\mathbb{C}$ and can not be a linear algebraic group. A more persuasive but nontrivial example, pointed out to us by a referee, is $\text{Aut } \mathbb{C}^3$: it is not generated by the maximal torus and root subgroups (even though there are infinitely many root subgroups). For instance, the famous Nagata automorphism

$$(x, y, z) \mapsto (x + (x^2 - yz)z, y + 2(x^2 - yz)x + (x^2 - yz)^2 z, z)$$

is so-called “wild”, i.e., it is not contained in the subgroup generated by the maximal torus and root subgroups [29].

1.5 Smoothness issues

Oda’s remark concerned the compactness assumption in Demazure’s theorem. However, historically, it is the smoothness assumption that has been a focal point in the literature. In fact, one novel point of the present article is to settle this issue.

Cox succeeded in extending the theorem to the orbifold case. In this article we completely remove the smoothness or orbifold assumptions of Demazure and Cox and present Theorem 1.1 for an *arbitrary* compact toric variety. In fact, except for one crucial step, Cox’s proof is insensitive to the orbifold assumption; see Proposition 3.12 and Remark 3.13 for more details on how we overcome this.

We also mention the master thesis of Bühler [7] cited by Nill [25, p. 122] and Bruns–Gubeladze [6, p. 716]. According to the latter citation, Bühler also claims to extend Demazure–Cox’s theorem to an arbitrary (not necessarily orbifold) compact toric variety [7, Satz 4.2]. Bühler’s proof [7, p. 10] carries over the error in Cox’s original article (See Remark 3.16)). Moreover, while [7, Satz 4.4] is equivalent (as a statement) to our Proposition 3.12, our proof of it is completely different: Bühler uses more combinatorial arguments to prove that the subvariety $V(\text{Ir}_\Delta)$ (Definition 3.8) is invariant under any degree-preserving automorphism, while we use more algebraic arguments to prove a more direct statement that the irrelevant ideal Ir_Δ is itself invariant.

1.6 Other extensions

Demazure’s original arguments have been generalized to establish that the automorphism group of a complete (the analog of compact, see Theorem 2.9) toric variety over an *arbitrary field* is a

smooth group-scheme spanned by the torus T and one-parameter subgroups corresponding to the roots [22]. The proof in op. cit. is not self-contained and relies on the original Demazure argument, as well as other heavy scheme machinery. It is worth mentioning that $\text{Aut } X$ automatically admits a scheme structure in the cases covered by Cox's result, since a linear algebraic group corresponds to a group-scheme naturally and is smooth as long as the base field has characteristic 0.

In a different direction, Bruns–Gubeladze studied the automorphism group of a toric variety by a different method. Their results are, in a sense, parallel to Cox's: contrary to Theorem 1.1, they require the variety to be *projective*, but they allow the variety to be arbitrarily singular, in particular, more singular than orbifold, over *any algebraically closed field* [6, Theorem 5.4].

The projectivity allows them to use a very ample polarization and the associated moment polytope to describe the automorphism group. They use a different method to construct the toric variety, viewing it as $\text{Proj } S_P$, where S_P is a graded ring determined by a fixed (very ample) polytope P (more precisely, P determines a very ample line bundle on X , and S_P is the corresponding projective coordinate ring). In comparison, Cox uses the information of the fan Δ (Definition 2.3) to construct his total coordinates ring and describe the automorphism group, without any reference to the polytope that defines the toric variety (which is absent if the variety is not projective).

However, these two results [6, 22] do not seem to imply Theorem 3.14 without additional work, as their stated results are less granular: they do not mention the semi-simple roots and reductivity, as in Cox's work.

Another extension of Cox's result is to the case of a variety with a torus action of complexity one (i.e., a $(\mathbb{C}^*)^{n-1}$ -action) [1]. That paper assumes X is non-toric because if X is toric, the explicit argument and statement would be different (for example, the combinatorial definition of roots will be different). Therefore, this result does not imply Theorem 1.1 either. Finally, Cox's homogeneous coordinate ring can also be used to study automorphism groups in the more general case of spherical varieties [5].

2 Preliminaries

This section defines the basics required for the proof of Theorem 3.14.

2.1 Cones, fans, and associated toric varieties

This subsection introduce the definition of fans and cones, and how they give rise to a toric variety.

Definition 2.1. [15, p. 3] An n -dimensional complex (analytic or algebraic) variety X is called *toric* if it contains a complex torus $(\mathbb{C}^*)^n$ as an open dense subset, such that the multiplication-action of the torus on itself extends to the whole variety. If X is smooth, it is a *toric manifold*.

There is a natural way to construct toric varieties by the following way.

Consider a lattice N of rank n and its dual lattice

$$M := N^* := \text{Hom}(N, \mathbb{Z}).$$

Denote the corresponding $\mathbb{R}$ -vector space and its dual (both isomorphic to $\mathbb{R}^n$)

$$N_{\mathbb{R}} := N \otimes_{\mathbb{Z}} \mathbb{R}, \quad M_{\mathbb{R}} := M \otimes_{\mathbb{Z}} \mathbb{R} = N_{\mathbb{R}}^*.$$

Definition 2.2. • A *rational convex polyhedral cone* (which is called a *cone* for short throughout this article) in $N_{\mathbb{R}}$ is a subset

$$\sigma = \sigma(v_1, \dots, v_d) := \left\{ \sum_{i=1}^d a_i v_i : a_i \geq 0, v_i \in N \right\}.$$

The cone σ is said to *be generated by* $v_1, \dots, v_d$.

- A *face* of a cone σ is an intersection of σ with a supporting hyperplane.
- A cone is called *strongly convex* if $\sigma \cap -\sigma = \{0\}$ [15, p. 14].

Definition 2.3. A *fan*

$$\Delta := \{\sigma_i\}_{i=1}^{\mathcal{D}}$$

in N is a finite set of rational strongly convex polyhedral cones

$$\sigma_i \subset N_{\mathbb{R}}$$

satisfying:

- (i) each face of a cone in Δ is also a cone in Δ ,
- (ii) the intersection of two cones in Δ is a face of each.

Definition 2.4. A fan Δ defines a toric variety

$$X_{\Delta}$$

as follows: Each cone σ_i in Δ has a dual cone in $M_{\mathbb{R}}$

$$\sigma_i^* = \{m \in M_{\mathbb{R}} : \langle m, v \rangle \geq 0, \forall v \in \sigma_i\}$$

which gives rise to an affine toric variety $\text{Spec} \mathbb{C}[\sigma_i^* \cap M]$ [15, §1.3]. These affine varieties serve as a (Zariski open) chart in X_{Δ} . Gluing them by the transitions between the charts constructed by (i) and (ii) above, we obtain the toric variety X_{Δ} , which is called *the toric variety defined by Δ* . For more detail, see [15, p. 21].

Remark 2.5. The zero cone corresponds to the open dense orbit $T_X \cong (\mathbb{C}^*)^n$ [8, p. 64], and more generally, there is a bijection between the cones $\{\sigma_i\}_{i=1}^{\mathcal{D}}$ and the closure of the orbits of the torus action in X_{Δ} [8, Proposition 5.6.2], with the non-zero cones corresponding precisely to all the closed toric subvarieties of X_{Δ} of positive codimension. In particular, every 1-dimensional cone $\mathbb{R}^+v$ corresponds to a (toric invariant) divisor on X_{Δ} , which we denote by

$$v \leftrightarrow D_v.$$

By the construction of X (Definition 2.4), every point in $X \setminus T_X$ is contained in the closure of the toric orbit corresponding to a cone σ of positive dimension, thus in the closure of the toric orbit corresponding to any 1-dimensional face $\mathbb{R}^+v$ of σ . Hence, we have $X = T_X \cup (\bigcup D_v)$.

T_X acts naturally on itself, and this toric action extends to X . We can view M as the character group of T_X . Namely

$$M = \{m : T_X \rightarrow \mathbb{C}^* : m \text{ is a morphism and a group homomorphism}\}.$$

Remark 2.6. In Demazure’s original paper [12], he used the term “éventail” [12, Definition 1, p. 557] to define the toric variety over $\mathbb{Z}$ [12, Definition 2, p. 558]. See [11, Appendix A] for a discussion on how this corresponds to the more modern terminology of “fan”. Here is the gist. The “éventail” Demazure used is equivalent to a fan that we use today when we use it to define the toric variety. When one defines a fan, one gets a collection of (rational) cones $\{\sigma_i\}$. The corresponding éventail is the collection of sets of lattice points $\{\sigma_i \cap M^*\}$. One may obtain a fan from an éventail by considering the cones spanned by these sets of lattice points. Hence, they are equivalent to each other.

2.2 Regular, simplicial, and complete fans

Here are some basic terminologies about cones and fans:

Definition 2.7. A cone is

- *smooth* (or *regular*) if it is generated by a subset of a basis of N .
- *simplicial* if it is generated by a subset of a basis of $N_{\mathbb{R}}$.

A fan is

- *smooth* (or *regular*) if every cone in it is smooth (regular).
- *simplicial* if every cone in it is simplicial.
- *complete* if the support of it (i.e., the union of all the cones in it) is the whole $N_{\mathbb{R}}$.

Definition 2.8. [2, Definition 1.27, p. 14] (cf. [11, §11.4]) A (complex) *orbifold* is a Hausdorff topological space X with an atlas of charts

$$\{(U_i, \tilde{U}_i, \Gamma_i, \pi_i)\}$$

such that:

- $\{U_i\}$ is an open cover of X .
- $\tilde{U}_i \subseteq \mathbb{C}^n$ is a connected open subset.
- Γ_i is a *finite* group of biholomorphisms of $\tilde{U}_i$.
- $\pi_i : \tilde{U}_i \rightarrow U_i$ induces a homeomorphism $U_i \cong \tilde{U}_i / \Gamma_i$.
- the intersection of two charts U_i, U_j is again a chart U_k in the open cover $\{U_i\}$.
- for each inclusion $U_i \subseteq U_j$ there is an embedding $\Gamma_i \hookrightarrow \Gamma_j$ such that the inclusion $U_i \hookrightarrow U_j$ can be uniquely (up to a composition of group elements in Γ_i) lifted to a Γ_i -equivariant holomorphism $\tilde{U}_i \rightarrow \tilde{U}_j$.

Theorem 2.9. [11, Theorem 3.1.19, Theorem 3.4.6] *Let $X = X_{\Delta}$ be the toric variety (Definition 2.1) defined by a fan Δ (Definition 2.4), then:*

- X is compact (with respect to the complex analytic topology, equivalently, proper under the Zariski topology) if and only if Δ is complete (Definition 2.7).*
- X is smooth (regular) if and only if Δ is smooth (regular).*
- X is an orbifold (Definition 2.8) if and only if Δ is simplicial (Definition 2.7).*

Definition 2.10. A toric variety is called

- complete or compact if the corresponding fan is complete;
- simplicial or an orbifold if the corresponding fan is simplicial.

Example 2.11. The weighted projective space $\mathbb{P}(1, 1, 2)$ is a complete, simplicial, but non-smooth toric variety.

For a cone σ , the corresponding affine toric variety X_{σ} is an incomplete toric variety.

Unless otherwise specified, all the toric variety and the fan mentioned in this article are complete.

2.3 Polytopes associated to fans

We now associate a polytope with a fan (and a toric variety).

Definition 2.12. Let Δ be a complete fan, and let $\sigma_1, \dots, \sigma_d$ be its 1-dimensional cones (i.e., rays). Denote the *set of primitive vectors* $v_1, \dots, v_d$ that generate $\sigma_1, \dots, \sigma_d$ (i.e., $\sigma_i = \mathbb{R}_+ v_i$) by

$$\Delta_1 := \{v_1, \dots, v_d\} \subset N.$$

We may define:

Definition 2.13. The *polytope* P associated to Δ and X_Δ (Definition 2.4) is

$$P := \bigcap_{i=1}^d \left\{ y \in M_{\mathbb{R}} : \langle y, -v_i \rangle \leq 1 \right\} = \left\{ y \in M_{\mathbb{R}} : \max_j \langle -v_j, y \rangle \leq 1 \right\}. \quad (1)$$

Remark 2.14. • P contains the origin in its interior. P is automatically a polyhedron by definition, and is a polytope since we assume the fan is complete ([11, Proposition 4.3.8]).

- Equation (1) is the standard convention in most of the literature since then lattice points of P correspond to monomials (an exception is Batyrev–Selivanova that use $-P$ instead).
- P is exactly the moment (or Delzant) polytope associated to the polarization $(X, -K_X)$ when X is smooth and projective. In general, one can define such polytopes for any toric polarization. If the polarization used is $-K_X$ and $-K_X$ is *ample*, P recovers Δ (by polarity), $\Delta = (-P)^\circ$ (see [19]), and hence also $X = X_\Delta$ (Definition 2.4) itself. More on this in Remark 2.17.

The polytope P is important in this article, mainly since it encodes discrete information, roots, that ultimately characterizes $\text{Aut } X$. For some examples, refer to Figures 2 and 3.

Definition 2.15. The *roots* of a complete fan Δ and the toric variety X_Δ defined by Δ (sometimes called the roots of the associated polytope P (Definition 2.13)) are the elements of the set

$$R(\Delta) := \{m \in M : \exists v \in \Delta_1 \text{ such that } \langle m, v \rangle = -1 \text{ and } \langle m, v' \rangle \geq 0, \forall v' \neq v \in \Delta_1\}.$$

The set $R(\Delta)$ consists of two subsets:

$$R_s(\Delta) := -R(\Delta) \cap R(\Delta), \quad R_u := R(\Delta) \setminus R_s(\Delta).$$

The elements in $R_s(\Delta)$ are called the *semisimple roots* of Δ and X_Δ (and the associated polytope P).

Remark 2.16. (Geometric intuition behind $R(\Delta)$) The set of roots can be directly read off from the polytope P : $R(\Delta)$ is exactly the set of *lattice points contained in the relative interiors of the codimension-1 faces of P* . $\langle m, v \rangle \geq -1$ for all $v \in \Delta_1$ means m is on the boundary of P , and $\langle m, v_i \rangle = -1$ means that m is in the codimension-1 faces of P vertical to v_i . Hence, by Definition 2.15, any $m \in R(\Delta)$ is contained in exactly one codimension-1 face of P , thus in the relative interior of this face.

Remark 2.17. (The rôle of P) In general, the polytope defined in Definition 2.13 may not have good properties. The vertices of P may not even be the lattice points in M . Indeed, P is a lattice polytope (i.e., every vertex of P is a lattice point) if and only if X is a Gorenstein variety (i.e., the canonical divisor K_X of X is Cartier). See [11, Theorem 8.3.4]. Particularly, when X is smooth, P is always a lattice polytope.

Furthermore, the codimension-1 faces of P may not correspond one-to-one-to the rays in Δ . Let us illustrate with an example:

Consider the fan Δ_1 generated by the vectors

$$\{(1, 1), (-1, -1), (1, -1), (-1, 1), (1, 0)\}.$$

The associated polytope P_1 (Definition 2.13) is the same as the polytope associated to the fan Δ_2 generated by the subset

$$\{(1, 1), (-1, -1), (1, -1), (-1, 1)\}.$$

P_1 has only four codimension-1 faces, which can not correspond one-to-one to the five rays in Δ_1 . In fact, the codimension-1 faces of P correspond one-to-one to the rays in Δ if and only if X_Δ is $\mathbb{Q}$ -Fano (i.e., $-mK_X$ is Cartier and ample for some $m \in \mathbb{N}$), and in this case, the codimension-1 faces of P

$$\{y \in P : \langle y, -v_i \rangle = 1\}$$

are in one-to-one correspondence with the 1-dimensional cones generated by v_i in Δ , thus the irreducible toric divisors in X .

2.4 Automorphisms of toric varieties

In this subsection, Δ denotes a complete fan (Definition 2.3), $X := X_\Delta$ denotes the toric variety defined by Δ (Definition 2.4), and P denotes the polytope associated to the fan (Definition 2.13).

Definition 2.18. For a variety X , a morphism $f : X \rightarrow X$ is called an *automorphism* of X if f admits an inverse $f^{-1} : X \rightarrow X$ satisfying $f \circ f^{-1} = \text{id}_X$. The *automorphism group* of X , denoted by

$$\text{Aut } X,$$

is the group of all automorphisms of X . When X is proper over $\mathbb{C}$, $\text{Aut } X$ admitting a (complex) Lie group structure [23, Theorem 1.4], and the connected component of $\text{Aut } X$ that contains the identity is denoted by

$$\text{Aut}_0 X.$$

Remark 2.19. When X is a complex analytic variety, one may also define its complex analytic automorphism group (it is the group consists of all biholomorphism from X to X when X is a complex manifold).

It is worth noting that, for an arbitrary complex algebraic variety, its algebraic automorphism group (the automorphism group defined in Definition 2.18) may not be the same as the complex analytic automorphism group of the analytic variety associated to it. $\mathbb{C}^2$ is a simple example, since $(x, y) \mapsto (x, y + f(x))$ is a biholomorphism for any holomorphic function f but is algebraic automorphism for only polynomials f .

Nevertheless, when X is proper (compact in complex analytic topology), these two group are isomorphic to each other by generalized GAGA principle, see [17, Appendix B].

Definition 2.20. The subgroup of the automorphism group of the lattice M that leaves the polytope P (Definition 2.13) invariant is denoted by

$$\text{Aut } P \subseteq \text{GL}(M) \cong \text{GL}(n, \mathbb{Z}).$$

Similarly, The subgroup of the automorphism group of the lattice N that preserves the fan Δ (Definition 2.3) is denoted by

$$\text{Aut } \Delta \subseteq \text{GL}(N) \cong \text{GL}(n, \mathbb{Z}).$$

Remark 2.21. $\text{Aut } \Delta$ is a finite group since every element in it gives a permutation of Δ_1 (Definition 2.12) and this embeds $\text{Aut } \Delta$ into the permutation group of Δ_1 . Similarly, $\text{Aut } P$ is a finite group since every element in it gives a permutation of the vertices of P and this embeds $\text{Aut } P$ into a permutation group.

Remark 2.22. It is worth noting that the $\text{Aut } P$ we define above may be different from the subgroup of $\text{GL}(M \otimes \mathbb{R})$ that preserves the polytope P when P is not a lattice polytope. And so does $\text{Aut } \Delta$.

For example, if P is the convex hull of $\{(0, 1), (0, -1), (1/2, 0), (-1/2, 0)\}$, then linear transformation (multiply to the right)

$$\begin{pmatrix} 0 & 2 \\ 1/2 & 0 \end{pmatrix}$$

preserves P , but it is not in $\text{GL}(M)$.

When X is Fano, the dual map from $\text{GL}(N)$ to $\text{GL}(M)$ induces an isomorphism from $\text{Aut } \Delta$ to $\text{Aut } P$:

Lemma 2.23. *If X is Fano (i.e., $-K_X$ is Cartier and ample), then $\text{Aut } \Delta \cong \text{Aut } P$ (Definition 2.20).*

Proof. For an $\mathcal{P} \in \text{Aut } \Delta$, the dual of it is defined by

$$\mathcal{P}^* \in \text{GL}(M) : \langle m, \mathcal{P}n \rangle = \langle \mathcal{P}^*m, n \rangle, \forall m \in M, n \in N$$

For a $\mathcal{P} \in \text{Aut } \Delta$ and any $m \in P \cap M_{\mathbb{R}}, \sigma_i \in \Delta_1$, we have

$$\langle \mathcal{P}^*m, v_i \rangle = \langle m, \mathcal{P}v_i \rangle \geq -1$$

since $\mathcal{P}v_i$ is a primitive vector in Δ_1 . This implies that $\mathcal{P}^*m \in P$, which shows $\mathcal{P}^*$ preserves the polytope P and so $\mathcal{P}^* \in \text{Aut } P$.

Conversely, if $\mathcal{P}^* \in \text{Aut } P$, we want to show that $\mathcal{P}v_i \in \Delta_1$. Because $v_i \in \Delta_1$ and X is Fano, there is a codimension-1 face $F' \subseteq P$ that satisfies $\langle m, v_i \rangle = -1$ for all $m \in F' \cap M_{\mathbb{R}}$. Then we have

$$\langle m, \mathcal{P}v_i \rangle = \langle \mathcal{P}^*m, v_i \rangle = -1, \forall m \in (\mathcal{P}^*)^{-1}F'$$

And $(\mathcal{P}^*)^{-1}F'$ is another codimension-1 face of P since $\mathcal{P}^*$ preserves the polytope P . Thus, $\mathcal{P}v_i$ is another primitive vector in Δ_1 , which shows $\mathcal{P} \in \text{Aut } \Delta$. $\square$

There is a special subgroup of $\text{Aut } \Delta$ that will play an important role later.

Definition 2.24. For every $m \in R_u$, there is a unique pair $(v, v') \in \Delta_1 \times \Delta_1$ such that $\langle m, v \rangle = -\langle m, v' \rangle = 1$ and $\langle m, v_i \rangle$ for other $v_i \in \Delta_1$. Hence, m gives a reflection

$$\text{Aut } \Delta \ni \mathcal{P}_m := n \mapsto n + \langle m, n \rangle (v - v').$$

The subgroup of $\text{Aut } \Delta$ generated by all the $\mathcal{P}_m$ is denoted by

$$\text{Aut}_0 \Delta.$$

We will see that (in Proposition 4.25) this is exactly the subgroup of $\text{Aut } \Delta$ that induces the identity map on the divisor class group of X (Definition 3.1).

2.5 Reductive Lie groups and maximal tori

We will use some basic concepts and facts about Lie groups in this article. For the convenience of readers, we list them in this subsection without proof. Readers who are interested in the details can refer to standard textbooks on Lie groups, such as [16] and [18].

Definition 2.25. • A complex Lie group is a *linear algebraic group* if it can be embedded into $\mathrm{GL}(k, \mathbb{C})$ for some $k \in \mathbb{N}$ as a closed subgroup.

- A *faithful representation* of a group G is a representation (that is, a morphism from G to $\mathrm{GL}(V)$, where V is a linear space) with a trivial kernel.
- Let G be a connected linear algebraic group. An element $g \in G$ is called *unipotent* if for a faithful representation φ , $\varphi(g) - I$ is unipotent (i.e., $(\varphi(g) - I)^k = 0$ for some $k > 0$). A subgroup of G is called *unipotent* if every element of it is unipotent.
- The *unipotent radical* of a connected linear algebraic group G is the largest connected unipotent normal subgroup of G . G is called *reductive* if its unipotent radical is trivial.

Definition 2.26. • For a complex Lie group G , a *torus* in G is a Lie subgroup isomorphic to $(\mathbb{C}^*)^k$. All the tori in G are partially ordered by the inclusion relation. A maximal element under this partial order is called a *maximal torus* of G .

- Let T be a maximal torus in a (connected) reductive complex Lie group G , and $N(T)$ be its normalizer. Then we call

$$W(T) := N(T)/T$$

the *Weyl group* of G with respect to the torus T .

Example 2.27. As a simple example, if $G = \mathrm{GL}(n, \mathbb{C})$, we can choose a maximal torus T formed by all diagonal matrices. With respect to this torus, the Weyl group W is the permutation group S_n of the n coordinates. Hence, W can be identified with the subgroup of G consisting of all permutation matrices naturally.

Theorem 2.28. [16, Theorem 11.9] *Let G be a linear algebraic group (Definition 2.25), then all maximal tori (Definition 2.26) in G are conjugate to each other.*

Remark 2.29. In [16], the above theorem is proved for compact (real) Lie groups. However, combining the fact that any reductive complex Lie group can be obtained from a compact real Lie group by “complexification” (it is well known but nontrivial, see [18, Theorem 15.3.11] and [24, Theorem 3]) and the fact that the unipotent radical intersects with any maximal tori trivially, we can translate the results there to the above theorem.

3 The Structure Theorem

In this section, we first introduce Cox’s quotient construction of a toric variety as preparation for the proof of the Structure Theorem 3.14. Then, we give the precise statement of the Structure Theorem 3.14 and the basic idea of the proof using the quotient construction just introduced. In the last subsection, we provide some explicit examples to illustrate the Structure Theorem 3.14.

In this section, Δ denotes a complete fan (Definition 2.3), $X := X_\Delta$ denotes the toric variety defined by Δ (Definition 2.4), and P denotes the polytope associated to the fan (Definition 2.13).

3.1 The homogeneous coordinate ring

We first introduce the homogeneous coordinate ring defined by Cox, which will play a central role in the proof of Theorem 3.14.

Given a complete (Definition 2.10) toric variety $X = X_\Delta$ defined by a fan Δ (Definition 2.4), we define that:

Definition 3.1. The *Weil divisor group* $\text{Div}X$ of X is the free abelian group generated by all the prime Weil divisors on X . The principal divisors form a subgroup Div_0X of $\text{Div}X$, and the *Weil divisor class group* of X is the quotient group

$$\text{Cl}(X) := \text{Div}X / \text{Div}_0X.$$

Remark 3.2. If X is smooth, $\text{Cl}(X)$ is isomorphic to the Picard group of X . But for singular cases, $\text{Cl}(X)$ could be different from the Picard group. For example, if X is simplicial (Definition 2.10), then every Weil divisor has an integer multiple that is Cartier, and the Picard group is a subgroup of $\text{Cl}(X)$ with finite index. One of the simplest examples is the weighted projective space $\mathbb{P}(1, 1, 2)$: its divisor group is isomorphic to $\mathbb{Z}$, while its Picard group corresponds to the subgroup $2\mathbb{Z}$. For readers interested in the details, we refer to Cox's textbook [11, §4.0].

Definition 3.3. Denote the set of all primitive vectors generating the 1-dimensional cones in Δ by Δ_1 (Definition 2.12).

$$\mathbb{C}^{\Delta_1} := \bigoplus_{v \in \Delta_1} \mathbb{C}v$$

is the complex linear space spanned by the linearly independent basis Δ_1 , and

$$(\mathbb{C}^*)^{\Delta_1} := \bigoplus_{v \in \Delta_1} \mathbb{C}^*v$$

is a *maximal torus* in it.

Definition 3.4. [9, §1] Denote the set of all primitive vectors of 1-dimensional cones in Δ by Δ_1 (Definition 2.12). The *homogeneous coordinate ring* (which is commonly denoted by S)

$$\text{R}^{\text{HC}} := \mathbb{C}[\{x_v : v \in \Delta_1\}]$$

of $X = X_\Delta$ (Definition 2.4) is the polynomial ring over $\mathbb{C}$ generated by the variables $\{x_v : v \in \Delta_1\}$. Each $v \in \Delta_1$ corresponds to a divisor in X denoted by D_v (Remark 2.5). Hence, every monomial $\prod x_v^{a_v}$ in R^{HC} corresponds to a divisor $D = \sum a_v D_v$, thus a divisor class in $\text{Cl}(X)$. In this way, R^{HC} is graded by $\text{Cl}(X)$. For any $\alpha \in \text{Cl}(X)$, the *degree α subspace of R^{HC}* is denoted by

$$\text{R}_\alpha^{\text{HC}}.$$

3.2 The quotient construction of toric varieties

Cox uses the homogeneous coordinate ring R^{HC} (Definition 3.4) to construct a toric variety as a quotient space. In this subsection, we describe this construction (without proofs).

Definition 3.5. Divide $\{x_v : v \in \Delta_1\}$ into

$$\bigcup_{\alpha \in \text{Cl}(X)} \Delta_\alpha = \bigcup_{\alpha \in \text{Cl}(X)} \{x_v : (D_v) \in \alpha\}.$$

$\mathcal{C}$ is the set of divisor classes that contain at least one x_v , namely

$$\mathcal{C} := \{\alpha \in \text{Cl}(X) : \Delta_\alpha \neq \emptyset\}.$$

We define two groups that can act on $\mathbb{C}^{\Delta_1}$ (Definition 3.3) and X respectively.

Definition 3.6.

$$G := \text{Hom}_{\mathbb{Z}}(\text{Cl}(X), \mathbb{C}^*).$$

$$T'_X := \text{Hom}_{\mathbb{Z}}(M, \mathbb{C}^*).$$

By definition, T'_X corresponds to the action of the maximal torus in X , denoted by T_X , on X . We now show that G can act on $\mathbb{C}^{\Delta_1}$ naturally.

Lemma 3.7. *There is an exact sequence:*

$$1 \longrightarrow G \longrightarrow (\mathbb{C}^*)^{\Delta_1} \longrightarrow T'_X \longrightarrow 1, \quad (2)$$

where $(\mathbb{C}^*)^{\Delta_1}$ is defined in Definition 3.3.

Proof. There is a well known exact sequence (see, for example, [15, §3.4])

$$0 \longrightarrow M \longrightarrow \bigoplus_{v \in \Delta_1} \mathbb{Z}D_v \longrightarrow \text{Cl}(X) \longrightarrow 0,$$

where the map $M \rightarrow \bigoplus_{v \in \Delta_1} \mathbb{Z}D_v$ (Definition 2.5) is

$$m \mapsto \sum_{v \in \Delta_1} \langle m, v \rangle D_v.$$

Since $\sum_{v \in \Delta_1} \langle m, v \rangle D_v = 0$ if and only if $m = 0$ (recall that we always assume the fan Δ is complete (Definition 2.9)), this sequence is exact in M . By the definition of principal divisor and Weil divisor class group (Definition 3.1), this sequence is exact on $\bigoplus_{v \in \Delta_1} \mathbb{Z}D_v$. The exactness on $\text{Cl}(X)$ follows from the facts that $X = T_X \cup (\bigcup D_v)$ (see Remark 2.5) and $\text{Cl}(T_X) = 0$.

Applying the exact contravariant functor $\text{Hom}_{\mathbb{Z}}(\cdot, \mathbb{C}^*)$ to this exact sequence implies the lemma. $\square$

Definition 3.8. For each cone $\sigma \in \Delta$, let σ_1 denote the set of all primitive vectors generating the dimension-1 faces of σ (Definition 2.12), and $x^{\hat{\sigma}}$ be the monomial

$$x^{\hat{\sigma}} := \prod_{v \in \Delta_1 \setminus \sigma_1} x_v.$$

The *irrelevant ideal* $\text{Ir}_{\Delta} \subseteq \mathbb{R}^{\text{HC}}$ is

$$\text{Ir}_{\Delta} := \langle x^{\hat{\sigma}} : \sigma \in \Delta \rangle.$$

The following fact will be used in the proof of Theorem 3.14.

Lemma 3.9. *For any fan Δ that contains at least two cones (in particular, a complete fan Δ), the closed subset $V(\text{Ir}_{\Delta})$ defined by Ir_{Δ} (Definition 3.8) has codimension at least 2.*

Proof. Since Ir_{Δ} is generated by monomials, $V(\text{Ir}_{\Delta}) \subseteq \bigcup_{v \in \Delta_1} V(x_v)$. On the other hand, for any 1-dimensional cone $\mathbb{R}^+v$ in Δ , there exists a $\sigma \in \Delta$ such that $\mathbb{R}^+v \not\subseteq \sigma$. Hence, by Definition 3.8, the variable x_v cannot divide the monomial $x^{\hat{\sigma}}$. Therefore, $V(\text{Ir}_{\Delta})$ contains no coordinate plane $V(x_v)$, which shows that $\text{codim} V(\text{Ir}_{\Delta}) \geq 2$. $\square$

As shown in the exact sequence (2), G acts on the affine space $\mathbb{C}^{\Delta_1}$ as a subgroup of $(\mathbb{C}^*)^{\Delta_1}$, leaving $V(\text{Ir}_{\Delta})$ (the subset defined by the ideal Ir_{Δ}) invariant.

Definition 3.10. Let G be a group acting on a scheme $\tilde{X}$. X is the

- *categorical quotient* $\tilde{X}/G$ if there is a G -invariant morphism

$$\pi : \tilde{X} \rightarrow X,$$

such that every G -invariant morphism from $\tilde{X}$ to another scheme factors through π .

- *geometric quotient* $\tilde{X}/G$ if it is the categorical quotient $\tilde{X}/G$ and

$$x \leftrightarrow \pi^{-1}(x)$$

gives a one-to-one correspondence between the points in X and the G -orbits in $\tilde{X}$.

For more precise definitions and further discussion, we refer to [11, §5.0]

Theorem 3.11. [11, §5.2] *Let X be a toric variety determined by a fan Δ (Definition 2.3), G be the group in Definition 3.6, $V(\text{Ir}_\Delta)$ be the close subset subset of $\mathbb{C}^{\Delta_1}$ (Definition 3.3) defined by the ideal Ir_Δ (Definition 3.8). We denote $\mathbb{C}^{\Delta_1} \setminus V(\text{Ir}_\Delta)$ by $\tilde{X}$. Then:*

- (i) $V(\text{Ir}_\Delta)$ is invariant under the action of G .
- (ii) X is isomorphic to the categorical quotient $\tilde{X}/G$.
- (iii) X is simplicial (see Theorem 2.9) if and only if it is isomorphic to the geometric quotient $\tilde{X}/G$.

3.3 Removing Cox's orbifold assumption

As explained in Remark 3.13 below, the core part of removing Cox's orbifold assumption is showing that when Δ is complete, Ir_Δ (Definition 3.8) is actually intrinsically defined by the multi-graded structure of R^{HC} (Definition 3.4), i.e., it is independent of the choice of generators of R^{HC} . To be precise, we have the following.

Proposition 3.12. *If Δ is a complete fan (Definition 2.9) and $\phi : R^{\text{HC}} \rightarrow R^{\text{HC}}$ is a degree-preserving automorphism of R^{HC} (Definition 3.4), then $\phi(\text{Ir}_\Delta) = \text{Ir}_\Delta$ (Definition 3.8).*

Proof. Since ϕ is invertible, it suffices to prove that $\phi(\text{Ir}_\Delta) \subseteq \text{Ir}_\Delta$, or equivalently, for any cone $\sigma \in \Delta$, $\phi(x^{\hat{\sigma}}) \in \text{Ir}_\Delta$. Suppose ϕ is written as $x_{v_i} \mapsto \phi_i$ where each ϕ_i is a polynomial of the same degree as x_{v_i} , then

$$\phi(x^{\hat{\sigma}}) = x^{\hat{\sigma}} \cdot \prod_{v_i \in \Delta_1 \setminus \sigma_1} \frac{\phi_i}{x_{v_i}}.$$

We prove $\phi(x^{\hat{\sigma}}) \in \text{Ir}_\Delta$ by induction:

First, we claim that $x^{\hat{\sigma}} \frac{\phi_1}{x_{v_1}} \in \text{Ir}_\Delta$.

Without loss of generality, we assume $\phi_1 = \prod x_v^{a_v}$ is a monomial of the same degree as x_{v_1} . Since the divisor $\sum a_v D_v - D_{v_1}$ is a principal divisor ($\prod x_v^{a_v}$ and x_{v_1} have the same degree), there is a character $m \in M$ such that $(m) = \sum a_v D_v - D_{v_1}$. This means $\langle m, v_1 \rangle = -1$ and $a_v = \langle m, v \rangle \geq 0$ for $v \neq v_1$. Hence, the hyperplane H perpendicular to m divides $N \otimes \mathbb{R}$ into two parts, and one of them contains only one vector in Δ_1 , namely v_1 . By definition 3.8, the variable x_v divides $x^{\hat{\sigma}} \frac{\phi_1}{x_{v_1}}$ if and only if $v \neq v_1$ and $v \notin H \cap \sigma$.

Noting that $H \cap \sigma$ is a face of σ . Consider all the cones in Δ that contain $H \cap \sigma$, there must be at least one, denoted by σ' , that touches the same side of H as v_1 since Δ is complete. Since $\mathbb{R}v_1$ is the only ray in Δ on the same side of H as v_1 , σ' must contain v_1 as a face (See Figure 1). If a variable x_v divides $x^{\hat{\sigma}'}$, then $v \neq v_1$ (since v_1 is a face of σ'), and $v \notin H \cap \sigma$ (since $H \cap \sigma$ is a

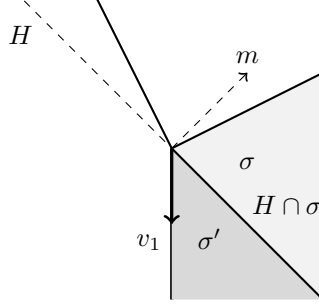


Figure 1: Finding the cone σ' in the proof of Proposition 3.12.

face of σ'). Hence, x_v must also divide $x^{\hat{\sigma}} \frac{\phi_1}{x_{v_1}}$. Therefore, $x^{\hat{\sigma}'}$ is a factor of $x^{\hat{\sigma}} \frac{\phi_1}{x_{v_1}}$, which implies $x^{\hat{\sigma}} \frac{\phi_1}{x_{v_1}} \in \text{Ir}_{\Delta}$ (Definition 3.8).

Next, since $x^{\hat{\sigma}} \frac{\phi_1}{x_{v_1}} \in \text{Ir}_{\Delta}$, it can be written as $\sum_j f_j x^{\sigma_j}$ for some polynomials f_j and cones σ_j . If x_{v_2} divides x^{σ_j} , then run the arguments above again, we obtain $f_j x^{\sigma_j} \frac{\phi_2}{x_{v_2}} \in \text{Ir}_{\Delta}$; otherwise, x_{v_2} must divide f_j , again we get that $f_j x^{\sigma_j} \frac{\phi_2}{x_{v_2}} \in \text{Ir}_{\Delta}$. Thus, $x^{\hat{\sigma}} \frac{\phi_1}{x_{v_1}} \frac{\phi_2}{x_{v_2}} \in \text{Ir}_{\Delta}$.

Therefore, by induction, we obtain that $\phi(x^{\hat{\sigma}}) \in \text{Ir}_{\Delta}$. $\square$

Remark 3.13. In other words, Proposition 3.12 proves that the “irrelevant part” $V(\text{Ir}_{\Delta})$ is invariant under any degree-preserving automorphism of $\mathbb{C}^{\Delta_1} = \text{SpecR}^{\text{HC}}$. Therefore, this lemma generalizes a statement in [9, Proposition 4.6] that $V(\text{Ir}_{\Delta})$ (denoted by Z there) is invariant under any degree-preserving automorphism of $\mathbb{C}^{\Delta_1} = \text{SpecR}^{\text{HC}}$ (denoted by an element of $\text{Aut}_g(S)$ there) if the fan is simplicial (noting that the other parts of [9, Proposition 4.6] did not use the simplicial condition at all). It is the only place in Cox’s proof of [9, Theorem 4.2] that essentially uses the simplicial condition. To be specific, Cox found a cone $\sigma' \in \Delta$ containing $H \cap \sigma$ and v_0 as faces. Then he used the simplicial condition to obtain that σ' must be spanned by $H \cap \sigma$ and v_0 , so that he could use this to prove that $\mathbb{C}^{\Delta_1} \setminus V(\text{Ir}_{\Delta})$ is invariant under any degree-preserving automorphism by analyzing the image of the points in $\mathbb{C}^{\Delta_1} \setminus V(\text{Ir}_{\Delta})$. Here we use a more algebraic argument in the ring and ideal level, which allows us to see clearly that we do not need a cone σ' spanned by $H \cap \sigma$ and v_0 . Instead, we only need a cone containing $H \cap \sigma$ and v_0 as faces. Thus, the simplicial condition is actually removable. As a consequence, Cox’s proof and result can be generalized to toric varieties with *arbitrary singularities*.

3.4 Statement and key idea of proof

The main task of the next section is to analyze the structure of the automorphism group $\text{Aut } X$ of X . We will eventually establish the following result, a more precise version of Theorem 1.1.

Theorem 3.14 (Demazure–Cox–Bühler Theorem). *Let $X = X_{\Delta}$ be a complete toric variety of dimension n (Definition 2.10) defined by a fan Δ (Definition 2.4), and P be the associated polytope (Definition 2.13). Then $\text{Aut } X$ (Definition 2.18) is a linear algebraic group that satisfies:*

- (i) $\text{Aut}_0 X$ (Definition 2.18) is a linear algebraic group (Definition 2.25) with a maximal torus $T'_X \cong (\mathbb{C}^*)^n$ (Definition 3.6).
- (ii) The unipotent radical (Definition 2.25) R_u of $\text{Aut}_0 X$ is generated by the one-parameter groups $\phi^{m, \lambda}$ (Definition 4.17) associated with the roots $m \in R_u(\Delta)$ (Definition 2.15). $\text{Aut}_0 X$

is the semi-direct product of R_u with a group G_s/G , where $G_s \cong \prod_{\alpha \in \mathcal{C}} \mathrm{GL}(|\Delta_\alpha|, \mathbb{C})$ (Definition 3.5) and G (given in Definition 3.6) is embedded in G_s as in the exact sequence (2).

- (iii) $\mathrm{Aut} \Delta$ (Definition 2.20) can be naturally embedded in $\mathrm{Aut} X$ such that $\mathrm{Aut} \Delta \cap \mathrm{Aut}_0 X \cong \mathrm{Aut}_0 \Delta$ (Definition 2.24), $\mathrm{Aut} X / \mathrm{Aut}_0 X \cong \mathrm{Aut} \Delta / \mathrm{Aut}_0 \Delta$ and $\mathrm{Aut}_0 \Delta$ is isomorphic to the Weyl group of G_s/G with respect to the maximal torus T'_X (Definition 2.26).

As a corollary, the following special case of geometric interest, referred to in §1.2 (and used in our companion paper [20]) as follows:

Corollary 3.15. *If X is complete (Definition 2.10), then P (Definition 2.13) is centrally lattice symmetric (i.e., $R(\Delta) = -R(\Delta)$ (Definition 2.15)) if and only if $\mathrm{Aut}_0 X$ is reductive (Definition 2.25).*

Proof. From part (ii) of Theorem 3.14, the unipotent radical of $\mathrm{Aut}_0 X$ is generated by the one-parameter groups associated with the roots in $R_u(\Delta)$. Therefore, $\mathrm{Aut}_0(X)$ is reductive (i.e., the unipotent radical of $\mathrm{Aut}_0(X)$ is trivial) if and only if $R_u(\Delta) = \emptyset$, which is equivalent to $R(\Delta) = -R(\Delta)$ since $R_u(\Delta) = R(\Delta) \setminus (-R(\Delta) \cap R(\Delta))$ (Definition 2.15). □

We now try to explain the basic idea of Cox’s proof of Theorem 3.14:

1. Since $\mathbb{C}^{\Delta_1} \setminus V(\mathrm{Ir}_\Delta)$ (Definition 3.3 and Definition 3.8) is simply an affine space minus some subvarieties cut out by coordinate functions, it is much easier to analyze its automorphism group. Therefore, we concentrate on the group of automorphisms of $\mathbb{C}^{\Delta_1} \setminus V(\mathrm{Ir}_\Delta)$ that can be descended to X , whose identity component turns out to consist of automorphisms given by some homogeneous polynomials with respect to the $\mathrm{Cl}(X)$ -grading of R^{HC} (Definition 3.4).
2. Observing that these homogeneous polynomials are determined by the degree-preserving linear endomorphism of R^{HC} , we can write them as matrices. This shows that they form a linear algebraic group and allows us to figure out the precise structure of this group.
3. To understand how the automorphism group $\mathrm{Aut} X$ is related to the combinatorial information of X , we study the relation between its unipotent radical R_u (Definition 4.11) and the roots $R(\Delta)$ (Definition 2.15) of Δ .
4. The next step is to describe the relation between the groups described above and $\mathrm{Aut} X$. Just as our intuition tells us, $\mathrm{Aut} X$ is the quotient group $\mathrm{Aut} X / G$ (Definition 4.1), but the proof requires effort. We first lift the automorphisms in the identity component of $\mathrm{Aut} X$ to $\mathrm{Aut}_0 X$ (Definition 4.6).
5. Finally, we show how the elements in $\mathrm{Aut} \Delta$ (Definition 2.20) “generate” other components of $\mathrm{Aut} X$.

Remark 3.16. In the next section, we provide a proof of Demazure’s Theorem mostly following Cox’s paper. We modify the proof from Cox’s original paper slightly to avoid some arguments. For example, the connectedness argument is used in the proof of Proposition 4.20 and the following corollary to replace some arguments in the original proof.

Cox’s original proof has a gap: he assumed that the automorphism group of a ring is an algebra (which is incorrect because the sum of two automorphisms is not a homomorphism of rings), so that he could run some arguments more easily. The gap is, of course, somewhat serious, but it was pointed out very clearly and fixed very well in his erratum [10].

3.5 Examples

To gain some intuition on how to prove the theorem (and also why it is true in the first place), it is insightful to look at the simplest examples: del Pezzo surfaces.

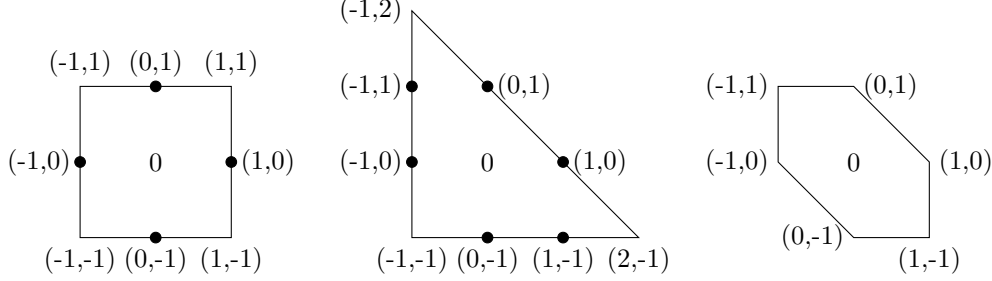


Figure 2: The polytope P corresponding to the toric del Pezzo surfaces with a *reductive* automorphism group: $\mathbb{P}^1 \times \mathbb{P}^1$, $\mathbb{P}^2$ and $\mathbb{P}^2$ blown up at three non colinear points. The points of $R(\Delta)$ are marked along the boundary of P . Notice that the automorphism group of P is isomorphic to the automorphism group of the polygon with ℓ vertices, i.e., $D_{2\ell} \subset S_\ell$, the dihedral group of order 2ℓ ; specifically, $\text{Aut } P \cong D_8, D_6 = S_3, D_{12}$, respectively, generated by cyclic permutations and reflections.

X	$\mathbb{P}^2$	$\mathbb{P}^1 \times \mathbb{P}^1$	$\mathbb{P}^2$ blown up 3 points
R^{HC}	$\mathbb{C}[x_0, x_1, x_2]$	$\mathbb{C}[x_0, x_1, y_0, y_1]$	$\mathbb{C}[x_0, x_1, x_2, y_1, y_2, y_3]$
G	$\mathbb{C}^*$	$(\mathbb{C}^*)^2$	$(\mathbb{C}^*)^4$
$\tilde{X}$	$\mathbb{C}^3 \setminus \{0\}$	$\mathbb{C}^4 \setminus (V(x_0, x_1) \cup V(y_0, y_1))$	$\mathbb{C}^6 \setminus (V(x_1 y_1 x_2 y_2) \cap V(y_1 x_2 y_2 x_3) \cap V(x_2 y_2 x_3 y_3) \cap V(y_2 x_3 y_3 x_1) \cap V(x_3 y_3 x_1 y_1) \cap V(y_3 x_1 y_1 x_2))$
$\text{Aut } X$	$\text{PGL}(3, \mathbb{C})$	$(\text{PGL}(2, \mathbb{C}) \times \text{PGL}(2, \mathbb{C})) \rtimes \mathbb{Z}/2\mathbb{Z}$	$(\mathbb{C}^*)^2 \rtimes D_{12}$
$\text{Aut}_0 X$	$\text{PGL}(3, \mathbb{C})$	$\text{PGL}(2, \mathbb{C}) \times \text{PGL}(2, \mathbb{C})$	$(\mathbb{C}^*)^2$
$\text{Aut } \Delta$	S_3	D_8	D_{12}
$\text{Aut}_0 \Delta$	S_3	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	0
R_u	0	0	0
G_s	$\text{GL}(3, \mathbb{C})$	$\text{GL}(2, \mathbb{C}) \times \text{GL}(2, \mathbb{C})$	$(\mathbb{C}^*)^6$

Table 1: Toric del Pezzo surfaces with reductive automorphism groups. In these cases $R_s(\Delta) = R(\Delta)$ and $R_u(\Delta) = \emptyset$, so $\text{Aut}_0 X$ is reductive. See Figure 2. Here $V(f_1, \dots, f_i)$ means the subvariety defined by the ideal generated by $f_1, \dots, f_i$.

Example 3.17. The simplest but nontrivial example may be $\mathbb{P}^2$. The homogeneous coordinate ring R^{HC} (Definition 3.4) of it is $\mathbb{C}[x_0, x_1, x_2]$, and the automorphism group is $\text{PGL}(3, \mathbb{C})$, which is known to be reductive. According to Figure 2, $\text{Aut } P = S_3$, and the roots of $\mathbb{P}^2$ are

$$\{(0, 1), (1, 1), (1, 0), (0, -1), (-1, -1), (-1, 0)\},$$

so the polytope of $\mathbb{P}^2$ is centrally lattice symmetric, although the polytope itself is not centrally symmetric.

Easily seen from the Lie algebra of $\text{PGL}(3, \mathbb{C})$, $\text{PGL}(3, \mathbb{C})$ is generated by the maximal torus

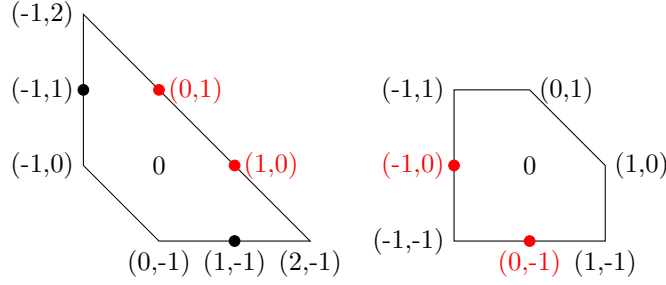


Figure 3: The polytope P corresponding to the toric del Pezzo surfaces with a *non-reductive* automorphism group: $\mathbb{P}^2$ blown up at one or two points. The points of $R(\Delta)$ are marked along the boundary of P , and the points in $R_u(\Delta)$ are marked in red. Since automorphisms of the lattice preserve the normalized volumes of faces, the only non-trivial automorphism of P is the reflection over the line $y = x$. Therefore, $\text{Aut } P \cong \mathbb{Z}/2\mathbb{Z}$, generated by $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

X	$\mathbb{P}^2$ blown up 1 point	$\mathbb{P}^2$ blown up 2 points
R^{HC}	$\mathbb{C}[x_0, x'_1, x'_2, y]$	$\mathbb{C}[x'_0, x'_1, x'_2, y_1, y_2]$
G	$(\mathbb{C}^*)^2$	$(\mathbb{C}^*)^3$
$\tilde{X}$	$\mathbb{C}^4 \setminus (V(x_0, y) \cup V(x'_1, x'_2))$	$\mathbb{C}^5 \setminus (V(x'_1, y_2) \cup V(x'_2, x'_0) \cup V(y_2, y_1) \cup V(x'_0, x'_1) \cup V(y_1, x'_2))$
$\text{Aut } X$	$\mathbb{C}^2 \rtimes \text{GL}(2, \mathbb{C})$	$(\mathbb{C}^2 \rtimes (\mathbb{C}^*)^2) \rtimes \mathbb{Z}/2\mathbb{Z}$
$\text{Aut}_0 X$	$\mathbb{C}^2 \rtimes \text{GL}(2, \mathbb{C})$	$\mathbb{C}^2 \rtimes (\mathbb{C}^*)^2$
$\text{Aut } \Delta$	$\mathbb{Z}/2\mathbb{Z}$	$\mathbb{Z}/2\mathbb{Z}$
$\text{Aut}_0 \Delta$	$\mathbb{Z}/2\mathbb{Z}$	0
R_u	$\mathbb{C}^2$	$\mathbb{C}^2$
G_s	$\text{GL}(3, \mathbb{C}) \times (\mathbb{C}^*)^2$	$(\mathbb{C}^*)^5$
$R_s(\Delta)$	$\{(1, -1), (-1, 1)\}$	$\emptyset$
$R_u(\Delta)$	$\{(1, 0), (0, 1)\}$	$\{(-1, 0), (0, -1)\}$

Table 2: Toric del Pezzo surfaces with non-reductive automorphism groups. See Figure 3. Here $V(f_1, \dots, f_i)$ means the subvariety defined by the ideal generated by $f_1, \dots, f_i$.

and the family below:

$$\begin{pmatrix} 1 & \lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & \lambda \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ \lambda & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \lambda \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \lambda & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \lambda & 1 \end{pmatrix}.$$

We will reveal the relation between them and the six roots of $\mathbb{P}^2$ in the next section.

Example 3.18. To see how the $\text{Cl}(X)$ -grading works, we blow up a 0-dimensional toric orbit, namely the intersection point x of D_1, D_2 , in $\mathbb{P}^2$. Then we get a ruled surface $X = \text{Bl}_x \mathbb{P}^2$, the homogeneous coordinate ring R^{HC} (Definition 3.4) of which is $\mathbb{C}[x_0, x'_1, x'_2, y]$. Here, y corresponds to the exceptional divisor E , and x'_i corresponds to the strict transform D'_i of D_i . As a ruled surface, the divisor class group of X is $\mathbb{Z} \oplus \mathbb{Z}$, where the first $\mathbb{Z}$ is the classes of fibers (D'_1 and D'_2 are here), the latter $\mathbb{Z}$ is the classes of sections (E is here), and D_0 is contained in the divisor class $(1, 1)$.

The automorphism group $\text{Aut } X$ can be viewed a subgroup of $\text{PGL}(3, \mathbb{C})$, generated by the maximal torus and the family below:

$$\begin{pmatrix} 1 & \lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & \lambda \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \lambda \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \lambda & 1 \end{pmatrix}.$$

This can be derived from the fact that the holomorphic vector fields on X (in other words, a vector in the Lie algebra of $\text{Aut } X$) are in one-to-one correspondence with the holomorphic vector fields on $\mathbb{P}^2$ that vanish at x . Equivalently, we can say that $\text{Aut } X$ contains the elements in $\text{PGL}(3, \mathbb{C})$ that preserve the point $x = [1, 0, 0]$ (very roughly, this is quite reasonable since E itself is the only prime divisor on X that is linearly equivalent to E , which forces every automorphism of X to preserve E).

Therefore, $\text{Aut } X$ is not reductive. The first two one-parameter groups above generate the unipotent radical of $\text{Aut } X$. The latter two one-parameter groups and the maximal torus in some sense generate the “reductive part” of $\text{Aut } X$, which is isomorphic to $\text{GL}(2, \mathbb{C})$.

According to Figure 2, $\text{Aut } P \cong \text{Aut } \Delta = \text{Aut}_0 \Delta = \mathbb{Z}/2\mathbb{Z}$, which again shows that $\text{Aut } X$ is connected in this case.

See Remark 4.15 for further discussion.

Example 3.19. We briefly introduce another simple example, $\mathbb{P}^1 \times \mathbb{P}^1$. In this case, the automorphism group is not connected, since, besides the identity component $\text{PGL}(2, \mathbb{C}) \times \text{PGL}(2, \mathbb{C})$ (it is a reductive group generated by the maximal torus and four one-parameter groups corresponding to the roots), there is another special automorphism P that exchanges the two $\mathbb{P}^1$'s. This automorphism has a distinctive feature that makes it so different from the automorphisms appearing above: it induces a nontrivial map on the divisor class group.

We can see this phenomenon in $\text{Aut } P$ directly: Here, $\text{Aut } \Delta \cong \text{Aut } P$ is the dihedral group D_8 , and $\text{Aut}_0 \Delta \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ is the subgroup generated by the up-down flip and the left-right flip. The quotient group is isomorphic to $\mathbb{Z}/2\mathbb{Z}$. Thus, $\text{Aut } X \cong (\text{PGL}(2, \mathbb{C}) \times \text{PGL}(2, \mathbb{C})) \rtimes \mathbb{Z}/2\mathbb{Z}$.

In other toric del Pezzos, $\text{Aut}_0 \Delta$ is equal to either the trivial group or $\text{Aut } \Delta$ itself.

Example 3.20. Suppose X is $\mathbb{P}^2$ blown up at two points. Then its homogeneous coordinate ring R^{HC} (Definition 3.4) is $\mathbb{C}[x'_0, x'_1, x'_2, y_1, y_2]$, where y_i corresponds to the exceptional divisors and x'_i corresponds to the strict transforms of the coordinate lines in $\mathbb{P}^2$.

Similar to Example 3.18, the identity component $\text{Aut}_0 X$ of the automorphism group of $\mathbb{P}^2$ blown up two points $[1, 0, 0], [0, 1, 0]$ is generated by the maximal torus $(\mathbb{C}^*)^2$ and the family:

$$\begin{pmatrix} 1 & 0 & \lambda \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \lambda \\ 0 & 0 & 1 \end{pmatrix},$$

since it must preserve the two points $[1, 0, 0], [0, 1, 0]$.

The difference is that there is another component, the elements of which exchange the points $[1, 0, 0], [0, 1, 0]$. We choose a typical member of this:

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

This matrix generates a subgroup $\mathbb{Z}/2\mathbb{Z}$. This corresponds to $\text{Aut } P$ according to Figure 3. Therefore, the automorphism group is a semi-direct product of $\mathbb{Z}/2\mathbb{Z}$ and $\text{Aut}_0 X$.

Example 3.21. The automorphism group of $\mathbb{P}^2$ with three non-colinear points blown up is isomorphic to the semi-direct product of $(\mathbb{C}^*)^2$ and the dihedral group D_{12} . Roughly speaking, the reason is as follows:

The three exceptional divisors and the strict transforms of the lines connecting the three points form a regular hexagon: the vertices of the hexagon correspond to these six divisors, and there is an edge connecting two vertices if and only if the divisors corresponding to them intersect. An automorphism of $\mathbb{P}^2$ with three nonlinear points blown up must preserve the set of these six divisors and their intersection relation. The subgroup of the automorphism group of $\mathbb{P}^2$ that preserves the three blown up points is the torus $(\mathbb{C}^*)^2$ itself. The above discussion tells us that $\text{Aut } X/(\mathbb{C}^*)^2$ can be identified as the automorphism group of the hexagon.

In fact, there is a natural subgroup $D_{12} \cong \text{Aut } P \subseteq \text{Aut } X$ that contains the automorphisms induced by the isomorphisms of the polytope (see Figure 2).

Keeping these examples in mind, we will figure out the structure of the automorphism groups of smooth toric varieties in the next section.

4 Proof of Theorem 3.14

In this section, Δ denotes a complete fan (Definition 2.3), $X := X_\Delta$ denotes the toric variety defined by Δ (Definition 2.4), and P denotes the polytope associated to the fan (Definition 2.13), $\tilde{X}$ denotes the space $\mathbb{C}^{\Delta_1} \setminus V(\text{Ir}_\Delta)$ in Theorem 3.11.

We break down the proof into manageable steps.

4.1 Step 1. Lifting automorphisms to $\tilde{X}$

According to the basic idea we mentioned in the last section, X can be viewed as a categorical quotient of $\tilde{X}$ by $G = \text{Hom}_{\mathbb{Z}}(\text{Cl}(X), \mathbb{C}^*)$ (see Theorem 3.11), and we shall first find the automorphisms of $\tilde{X}$ that can descend to X . Such automorphism ϕ should send one G -orbit to another. To be specific, for any $x \in \tilde{X}$ and $g \in G$, ϕ should satisfy

$$\phi(g \cdot x) \in G \cdot \phi(x), \quad (3)$$

which is equivalent to

$$\phi^{-1}G\phi = G.$$

This inspires us to define:

Definition 4.1.

$$\widetilde{\text{Aut } X}$$

is the *normalizer* of $G = \text{Hom}_{\mathbb{Z}}(\text{Cl}(X), \mathbb{C}^*)$ in $\text{Aut } \tilde{X}$, where $\tilde{X}$ denotes $\mathbb{C}^{\Delta_1} \setminus V(\text{Ir}_\Delta)$ (Definition 3.8).

The following lemma enables us to write down the elements in $\widetilde{\text{Aut } X}$ explicitly as polynomials, which we will use repeatedly from now on.

Lemma 4.2. $\widetilde{\text{Aut } X}$ is a subgroup of $\text{Aut } \mathbb{C}^{\Delta_1}$.

Proof. Since $V(\text{Ir}_\Delta)$ has codimension at least two (Lemma 3.9), every $\phi \in \text{Aut } \tilde{X}$ can be extended to an endomorphism of $\mathbb{C}^{\Delta_1}$ by the Riemann extension theorem. Thus, it is an automorphism by the Zariski-Nagata purity theorem, which asserts that, on a regular algebraic variety, the branch locus of a morphism must consist of pure codimension-1 subvarieties (see [31]). We get $\widetilde{\text{Aut } X} \subseteq \text{Aut } \tilde{X} \subseteq \text{Aut } \mathbb{C}^{\Delta_1}$. $\square$

The discussion above suggests that $\widetilde{\text{Aut}} X$ is exactly the group containing all the automorphisms of $\tilde{X}$ that can descend to X .

Lemma 4.3. *We have an exact sequence $1 \longrightarrow G \longrightarrow \widetilde{\text{Aut}} X \longrightarrow \text{Aut } X$, where G is the group defined in Definition 3.6.*

Proof. By (3), every $\phi \in \widetilde{\text{Aut}} X$ defines a G -invariant morphism

$$\pi \circ \phi : \tilde{X} \rightarrow X,$$

where π is the natural projection given by the definition of the categorical quotient (Definition 3.10). Hence, ϕ descends to an automorphism of X (see Theorem 3.11 and Definition 3.10). This provides the morphism $\widetilde{\text{Aut}} X \rightarrow \text{Aut } X$.

Every element in G descends to the identity of X since it preserves all the G -orbits. We show that every $\phi \in \widetilde{\text{Aut}} X$ that descends to the identity of X is contained in G :

By Theorem 3.11, over the smooth locus X_{reg} of X , the quotient $\tilde{X} \rightarrow X$ is geometric (actually, the action is locally free). Therefore, if $\phi \in \widetilde{\text{Aut}} X$ descends to the identity of X , it must preserve every G -orbit in $\pi^{-1}(X_{\text{reg}})$. Since X is normal, the codimension of $X \setminus X_{\text{reg}}$ is at least 2 and so is $\text{codim } \pi^{-1}(X \setminus X_{\text{reg}})$; thus, $V(x_v) \cap \pi^{-1}(X_{\text{reg}}) \neq \emptyset$ for the coordinate hyperplanes $V(x_v)$. Writing each component of ϕ as ϕ_v (Lemma 4.2), we claim that the polynomial ϕ_v cannot contain any variable except x_v . Otherwise, it would send some point outside the coordinate hyperplane $V(x_v)$ to $V(x_v) \cap \pi^{-1}(X_{\text{reg}})$, contradicting the fact that coordinate hyperplanes and $\pi^{-1}(X_{\text{reg}})$ are invariant under G . Furthermore, ϕ_v must have degree 1, otherwise, different points would be sent to one point by ϕ . Thus, $\phi \in (\mathbb{C}^*)^{\Delta^1}$. Now, the exact sequence (2)

$$1 \longrightarrow G \longrightarrow (\mathbb{C}^*)^{\Delta^1} \longrightarrow T'_X \longrightarrow 1$$

and the assumption that ϕ descends to the identity of X show that $\phi \in G$. $\square$

The proof of the fact that every automorphism of X can be lifted to $\widetilde{\text{Aut}} X$ (Definition 4.1) is more difficult and requires more preparation, so we will skip it temporarily and put it at the end of this section (see Proof of Proposition 4.4 in §4.5). Having admitted this, we have:

Proposition 4.4. *Let G be the group defined in Definition 3.6. Then*

$$1 \longrightarrow G \longrightarrow \widetilde{\text{Aut}} X \longrightarrow \text{Aut } X \longrightarrow 1$$

is an exact sequence.

Remark 4.5. When $X = \mathbb{P}^n$ this exact sequence is quite classical:

$$1 \longrightarrow \mathbb{C}^* \longrightarrow \text{GL}(n+1, \mathbb{C}) \longrightarrow \text{PGL}(n+1, \mathbb{C}) \longrightarrow 1.$$

When $X = \mathbb{P}^1 \times \mathbb{P}^1$, the exact sequence is

$$1 \longrightarrow (\mathbb{C}^*)^2 \longrightarrow \mathbb{Z}/2\mathbb{Z} \times (\text{GL}(2, \mathbb{C}))^2 \longrightarrow \mathbb{Z}/2\mathbb{Z} \times (\text{PGL}(n+1, \mathbb{C}))^2 \longrightarrow 1.$$

The group $G = (\mathbb{C}^*)^2$ is mapped into $(\text{GL}(2, \mathbb{C}))^2$ and forms the quotient group $(\text{PGL}(n+1, \mathbb{C}))^2$. The $\mathbb{Z}/2\mathbb{Z}$ in two groups both correspond to the automorphism that exchanges the two $\mathbb{P}^1$, thus canceling each other.

The next step is to understand the structure of $\widetilde{\text{Aut}} X$. It is natural to consider a smaller group formed by the automorphisms that commute with all the elements of G , since they look simply like homogeneous polynomials with respect to the $\text{Cl}(X)$ -grading.

Definition 4.6.

$$\widetilde{\text{Aut}}_0 X$$

is the *centralizer* of G in $\text{Aut } \widetilde{X}$.

4.2 Step 2. A faithful representation of $\widetilde{\text{Aut}}_0 X$ over $\oplus_{\alpha \in \mathcal{C}} \mathbb{R}_\alpha^{\text{HC}}$

In order to show that $\widetilde{\text{Aut}}_0 X$ is a linear algebraic group (Definition 2.25) over $\mathbb{C}$ and to analyze its structure, we construct a faithful representation (Definition 2.25) of it.

For each automorphism $\phi = (\phi_v)_{v \in \Delta_1}$ in $\widetilde{\text{Aut}}_0 X$, the map $x_v \mapsto \phi_v$ gives a $\mathbb{C}$ -algebra automorphism Φ^ϕ of $\mathbb{R}^{\text{HC}}$ (Definition 3.4). Formally speaking, ϕ is an automorphism of the affine variety $\mathbb{C}^{\Delta_1} \cong \text{Spec } \mathbb{R}^{\text{HC}}$, so it corresponds to an automorphism of $\mathbb{R}^{\text{HC}}$ naturally. We claim that:

Lemma 4.7. Φ^ϕ preserves the $\text{Cl}(X)$ -grading structure of $\mathbb{R}^{\text{HC}}$.

Proof. For $\phi \in \widetilde{\text{Aut}}_0 X$, write each component ϕ_v as (by Lemma 4.2)

$$\phi_v = \sum c_{v,D} x^D,$$

where $v \in \Delta_1$, D is a Weil divisor $D = \sum a_v D_v$, and

$$x^D := \prod x_v^{a_v}. \quad (4)$$

Then, for every $g \in G = \text{Hom}_{\mathbb{Z}}(\text{Cl}(X), \mathbb{C}^*)$

$$(\phi \circ g)_v = \sum g(D) c_{v,D} x^D, \quad (g \circ \phi)_v = g(D_v) \sum c_{v,D} x^D.$$

Hence, by the definition of G , every monomial x^D appearing in ϕ_v must satisfy

$$g(D) = g(D_v)$$

for every $g \in G$, thus having the same degree as x_v . The claim follows. $\square$

Therefore, we can restrict this automorphism to $\mathbb{R}_\alpha^{\text{HC}}$ for $\alpha \in \mathcal{C}$ (Definition 3.5), where $\mathbb{R}_\alpha^{\text{HC}}$ means the degree α part of the homogeneous coordinate ring $\mathbb{R}^{\text{HC}}$ (Definition 3.4), which provides a representation of $\widetilde{\text{Aut}}_0 X$ over the linear space $\oplus_{\alpha \in \mathcal{C}} \mathbb{R}_\alpha^{\text{HC}}$.

To be explicit:

Definition 4.8. The representation

$$\widetilde{\text{Aut}}_0 X \rightarrow \text{GL}(\oplus_{\alpha \in \mathcal{C}} \mathbb{R}_\alpha^{\text{HC}}) : \phi \mapsto \Phi^\phi \quad (5)$$

sends $\phi \in \widetilde{\text{Aut}}_0 X$ to a *block-diagonal matrix* Φ^ϕ , whose diagonal blocks are denoted by

$$\Phi_\alpha^\phi \in \text{GL}(\mathbb{R}_\alpha^{\text{HC}}).$$

Every row (and column) of Φ^ϕ corresponds to a monomial in $\mathbb{R}^{\text{HC}}$, thus a divisor D of the form $\sum a_v D_v$ for $v \in \Delta_1$. So, an entry of Φ^ϕ is positioned by two such divisors (D, D') .

For $D = \sum a_v D_v$ and $D' = \sum a'_v D_v$ in the same divisor class α , the (D, D') -entry

$$\varphi_{D, D'}$$

of the block Φ_α^ϕ is the coefficient of term $x^{D'}$ (see (4)) in the polynomial

$$\phi^D := \prod \phi_v^{a_v}. \quad (6)$$

Observing that the usual $\mathbb{Z}$ -degree of the nonzero term in ϕ^D can not be lower than that of x^D , under the natural monomial basis $\{x^D : \deg(D) = \alpha\}$ of R_α^{HC} (partially ordered by the usual $\mathbb{Z}$ -grading), the block Φ_α^ϕ looks like

$$\Phi_\alpha^\phi = \begin{pmatrix} A_\alpha & B_\alpha \\ 0 & C_\alpha \end{pmatrix}, \quad (7)$$

where A_α and B_α correspond to coefficients of ϕ_v for $v \in \Delta_\alpha$ (Definition 3.5) and the order of A_α is the cardinality $|\Delta_\alpha|$.

By Definition 4.8, the representation (5) is a faithful representation. Now we determine the image of the representation (5).

Definition 4.9.

$$\text{Aut}_g(R^{\text{HC}})$$

is the closed subset of $\text{GL}(\oplus_{\alpha \in \mathcal{C}} R_\alpha^{\text{HC}})$ defined by the following equations:

The entries of a matrix Φ in $\text{GL}(\oplus_{\alpha \in \mathcal{C}} R_\alpha^{\text{HC}})$ are positioned by two such divisors (D, D') as in Definition 4.8, denoted by $\varphi_{D, D'}$. The (D, D') -entry $\varphi_{D, D'} = 0$ if D and D' are not in the same divisor class α (Definition 3.1). Divide every block $\Phi_\alpha \in \text{GL}(R_\alpha^{\text{HC}})$ into four blocks as in (7). $\varphi_{D, D'} = 0$ for those entries not in the position of $A_\alpha, B_\alpha, C_\alpha$ in (7); for any entry $\varphi_{D, D'}$ in C_α , the equation is that $\varphi_{D, D'}$ equals a polynomial of some entries in A_α, B_α determined by the coefficient of term $x^{D'}$ in $\phi^D = \prod \phi_v^{a_v}$ (see Definition 4.8).

Lemma 4.10. $\text{Aut}_g(R^{\text{HC}})$ (Definition 4.9) is the image of $\widetilde{\text{Aut}}_0 X$ (Definition 4.6) under the representation (5). Moreover, $\text{Aut}_g(R^{\text{HC}})$ is a linear algebraic group (Definition 2.25) of dimension $\sum |\Delta_\alpha| \dim R_\alpha^{\text{HC}}$, where R_α^{HC} means the degree α part of the homogeneous coordinate ring R^{HC} (Definition 3.4).

Proof. By Definition 4.8, the image of an automorphism in $\widetilde{\text{Aut}}_0 X$ under the representation (5) is contained in $\text{Aut}_g(R^{\text{HC}})$. On the other hand, every matrix in $\text{Aut}_g(R^{\text{HC}})$ gives a degree-preserving automorphism of $\mathbb{C}^{\Delta_1}$, thus gives an automorphism of $\mathbb{C}^{\Delta_1}$ that preserves $V(\text{Ir}_\Delta)$ by proposition 3.12. Hence, it is the image of an automorphism in $\widetilde{\text{Aut}}_0 X$ under the representation (5). Therefore, the image of representation (5) is exactly the subset $\text{Aut}_g(R^{\text{HC}})$. As a consequence, $\text{Aut}_g(R^{\text{HC}})$ is a linear algebraic subgroup of $\text{GL}(\oplus_{\alpha \in \mathcal{C}} R_\alpha^{\text{HC}})$ (Definition 2.25).

Now, we compute the dimension of $\text{Aut}_g(R^{\text{HC}})$. First, note that the entries of all A_α, B_α are independent of each other, and the entries in C_α are determined by them. To make sure that Φ^ϕ is invertible, A_α, C_α must be invertible. However, it is an open condition (i.e., stable under small perturbations) and so it does not affect the dimension of $\widetilde{\text{Aut}}_0 X$. By Definition 4.8, $\dim A_\alpha + \dim B_\alpha = |\Delta_\alpha| \dim R_\alpha^{\text{HC}}$. Thus, we know that

$$\dim \widetilde{\text{Aut}}_0 X = \sum |\Delta_\alpha| \dim R_\alpha^{\text{HC}}.$$

□

From now on, we identify $\widetilde{\text{Aut}}_0 X$ with its image $\text{Aut}_g(\mathbb{R}^{\text{HC}})$ under the representation (5).

Now we can find the unipotent radical (see Definition 2.25) of $\widetilde{\text{Aut}}_0 X$. The unipotent radical of $\widetilde{\text{Aut}}_0 X$ should, in some sense, represent the influence of B_α , which inspires us to define:

Definition 4.11.

•

$$G_s$$

is the subgroup of $\widetilde{\text{Aut}}_0 X$ consisting of elements with diagonal blocks of the form

$$\begin{pmatrix} A_\alpha & 0 \\ 0 & C_\alpha \end{pmatrix}.$$

•

$$R_u$$

is the subgroup of $\widetilde{\text{Aut}}_0 X$ consisting of elements with diagonal blocks of the form

$$\begin{pmatrix} I & B_\alpha \\ 0 & C_\alpha \end{pmatrix}.$$

We claim:

Lemma 4.12. R_u (Definition 4.11) is the unipotent radical (Definition 2.25) of $\widetilde{\text{Aut}}_0 X$ (Definition 4.6) and there is a split exact sequence

$$1 \longrightarrow R_u \longrightarrow \widetilde{\text{Aut}}_0 X \longrightarrow \prod_{\alpha \in \mathcal{C}} \text{GL}(|\Delta_\alpha|, \mathbb{C}) \longrightarrow 1.$$

Proof. First, for diagonal block Φ_α^ϕ of any elements Φ^ϕ in R_u , C_α are upper triangular matrices with all eigenvalues 1. This is deduced by the definition of the representation (Definition 4.8) and noting that the components of ϕ look like

$$\phi_v = x_v + \text{higer-order terms}$$

and (see (6))

$$\phi^D = \prod \phi_v^{a_v} = x^D + \text{higer-order terms}.$$

Thus, all the elements in R_u are unipotent.

Next, Take an automorphism ϕ and write Φ^ϕ as in (7). Suppose $B_\alpha = 0$ for all α . Then Φ^ϕ is totally determined by A_α . Furthermore, every Φ^ϕ of this type has an inverse of the same type because, in this case, ϕ is just an invertible linear automorphism over $\mathbb{C}^{\Delta_1}$. So, Φ^ϕ 's of this type form a subgroup of $\widetilde{\text{Aut}}_0 X$, and this subgroup is reductive since it is isomorphic to $\prod_{\alpha \in \mathcal{C}} \text{GL}(|\Delta_\alpha|, \mathbb{C})$. we can identify $\prod_{\alpha \in \mathcal{C}} \text{GL}(|\Delta_\alpha|, \mathbb{C})$ with the subgroup G_s of $\widetilde{\text{Aut}}_0 X$ and describe the map from $\widetilde{\text{Aut}}_0 X$ to $\prod_{\alpha \in \mathcal{C}} \text{GL}(|\Delta_\alpha|, \mathbb{C})$ by keeping A_α , erasing B_α and recalculating C_α by the new A_α and B_α . By definition, this is a surjective group homomorphism, and the kernel is clearly R_u . Our claim follows from this. $\square$

Corollary 4.13. $\widetilde{\text{Aut}}_0 X$ is connected.

Proof. Both R_u and G_s are connected. Hence, the corollary follows from Lemma 4.12. $\square$

By now, we have shown that:

Proposition 4.14. $\widetilde{\text{Aut}}_0 X$ (Definition 4.6) is a connected linear algebraic group (Definition 2.25) of dimension $\sum |\Delta_\alpha| \dim R_\alpha^{\text{HC}}$ (Definition 3.5). Moreover, $\widetilde{\text{Aut}}_0 X$ is the semi-direct product of its unipotent radical R_u (Definition 4.11) with a reductive group $G_s \cong \prod_{\alpha \in \mathcal{C}} \text{GL}(|\Delta_\alpha|, \mathbb{C})$. $(\mathbb{C}^*)^{\Delta_1}$ (Definition 3.3) is a maximal torus of G_s , and thus a maximal torus of $\text{Aut}_0 X$.

Proof. By Lemma 4.10, Lemma 4.12, and Corollary 4.13. $\square$

Remark 4.15. Now we go back to Example 3.18, namely $\mathbb{P}^2$ blown up at a point. Here $\mathcal{C}$ (Definition 3.5) contains three classes, $(1, 0)$, $(0, 1)$ and $(1, 1)$. The linear spaces $R_{(0,1)}^{\text{HC}}$ and $R_{(1,0)}^{\text{HC}}$ are spanned by monomials y and $\{x'_1, x'_2\}$ respectively. The linear space $R_{(1,1)}^{\text{HC}}$ is spanned by monomials $\{x_0, yx'_1, yx'_2\}$. So, the matrices in $\widetilde{\text{Aut}}_0 X$ look like

$$\begin{pmatrix} \Phi_{(0,1)} & 0 & 0 \\ 0 & \Phi_{(1,0)} & 0 \\ 0 & 0 & \Phi_{(1,1)} \end{pmatrix},$$

where $\Phi_{(0,1)}$ is a nonzero complex number a_y , and $\Phi_{(1,0)}$ is a matrix $\begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix}$ in $\text{GL}(2, \mathbb{C})$. $\Phi_{(1,1)}$ is of the form

$$\begin{pmatrix} a_{0,0} & b_{0,1} & b_{0,2} \\ 0 & a_y a_{1,1} & a_y a_{1,2} \\ 0 & a_y a_{2,1} & a_y a_{2,2} \end{pmatrix}.$$

We can define an isomorphism from $\text{Aut}_0 X$ to a subgroup G' of $\mathbb{C}^* \times \text{GL}(3, \mathbb{C})$ by sending such a matrix to

$$a_y \times \begin{pmatrix} a_{0,0} & b_{0,1} & b_{0,2} \\ 0 & a_y a_{1,1} & a_y a_{1,2} \\ 0 & a_y a_{2,1} & a_y a_{2,2} \end{pmatrix}.$$

An element (g_1, g_2) in $G = (\mathbb{C}^*)^2$ corresponds to

$$\begin{pmatrix} g_2 I_1 & 0 & 0 \\ 0 & g_1 I_2 & 0 \\ 0 & 0 & g_1 g_2 I_3 \end{pmatrix}$$

in $\widetilde{\text{Aut}}_0 X$ (here I_i means the identity matrix of order i) and $g_2 \times (g_1 g_2 I_3)$ in $\mathbb{C}^* \times \text{GL}(3, \mathbb{C})$. In this way, the quotient $G'/(\mathbb{C}^*)^2$ gives the automorphism group that we have provided in Example 3.18.

4.3 Step 3. The relation between R_u and $R(\Delta)$

We can say more about the relation between $\widetilde{\text{Aut}}_0 X$, R_u , and the polytope P of X , or more precisely, the roots $R(\Delta)$ (see Definition 2.15).

Lemma 4.16. *There are one-to-one correspondences:*

$$\begin{aligned} R(\Delta) &\leftrightarrow \{(x_v, x^D) : \deg(x^D) = \deg(x_v), x_v \neq x^D\}, \\ R_s(\Delta) &\leftrightarrow \{(x_v, x_{v'}) : \deg(x'_{v'}) = \deg(x_v), x_v \neq x_{v'}\}. \end{aligned}$$

Here the degree means the $\text{Cl}(X)$ -grading (Definition 3.4) and x^D is as in (4).

Proof. According to the representation constructed above (Definition 4.8) and Proposition 4.14, whether $\widetilde{\text{Aut}}_0 X$ is reductive is equivalent to whether the orders of all C_α in the block Φ_α^ϕ (see (7)) are all 0, which is further equivalent to whether there exists a non-prime effective divisor D that is linearly equivalent to some prime divisor D_v .

Now suppose $D = \sum a_v D_v$ is such a divisor, then we can find a character $m \in M$ such that $(m) = D - D_{v_0}$ for some $v_0 \in \Delta_1$. This means that for $v \in \Delta_1$,

$$\langle m, v_0 \rangle = -1, \quad \langle m, v \rangle \geq 0 \text{ for } v \neq v_0.$$

The first equation shows that m is in the codimension-1 face, denoted by P_{v_0} , of P perpendicular to v_0 , and the second equation shows that m is in the relative interior of this face, therefore $m \in R(\Delta)$.

Conversely, for $m \in R(\Delta) \cap P_{v_0}$, $(m) = D - D_{v_0}$ gives an effective divisor D linearly equivalent to D_{v_0} . Furthermore, D is another prime divisor $D_{v'_0}$ which is equivalent to $-m \in R(\Delta) \cap P_{v'_0}$ since $(-m) = D_{v_0} - D$. $\square$

Just as the name “root” hints, similar to usual cases, the roots and the maximal torus can generate the whole group in the following sense:

Definition 4.17. For every $m \in R(\Delta)$, suppose $(m) = D - D_{v_0}$, associate it with a one-parameter subgroup $\phi^{m,\lambda} \subseteq \widetilde{\text{Aut}}_0 X$ named *root automorphism group* by setting

$$\phi_{v_0}^{m,\lambda} := x_{v_0} + \lambda x^D, \quad \phi_v^{m,\lambda} := x_v \text{ for } v \neq v_0.$$

When $m \in R_s(\Delta)$, $\phi^{m,\lambda}$ lies in G_s (see Proposition 4.14), otherwise $\phi^{m,\lambda}$ lies in R_u (Definition 4.11). Then we claim:

Proposition 4.18. $\widetilde{\text{Aut}}_0 X$ (Definition 4.6) is generated by the maximal torus $(\mathbb{C}^*)^{\Delta(1)}$ (Definition 3.3) and the one-parameter subgroups $\phi^{m,\lambda}$ for $m \in R(\Delta)$ (Definition 4.17).

Proof. For $m \in R_s(\Delta)$, $\phi^{m,\lambda}$ corresponds to the usual roots of the reductive group

$$G_s \cong \prod_{\alpha \in \mathcal{C}} \text{GL}(|\Delta_\alpha|, \mathbb{C}),$$

(namely, the matrices with 1 in all diagonal positions, $\lambda \in \mathbb{C}^*$ in another position, and 0 in other positions). They together with the maximal torus $(\mathbb{C}^*)^{\Delta_1}$ generate G_s , since the tangent vectors of these one-parameter groups and the maximal torus in the identity of $\widetilde{\text{Aut}}_0 X$ span its Lie algebra.

For $m \in R_u(\Delta)$, taking the derivative of $\Phi^{m,\lambda}$ (the image of $\phi^{m,\lambda}$ under the representation in Definition 4.8) with respect to λ , we get vectors

$$\theta^m := \left. \frac{d}{d\lambda} \right|_{\lambda=0} \Phi^{m,\lambda}$$

in the Lie algebra of R_u whose diagonal blocks are of the form

$$\theta_\alpha^m = \begin{pmatrix} 0 & E^m \\ 0 & C'_\alpha \end{pmatrix},$$

where C'_α is the derivative in the corresponding place, E^m has 1 in the (v_0, D) position, and 0 in other positions if $(m) = D - D_{v_0}$. Since x^D does not contain x^{v_0} , $\Phi^{m,\lambda}$ has 1 in the (D, D) -position and 0 in other positions in the D -row, which shows that the D -row of C'_α is 0 and for $k > 1$

$$(\theta_\alpha^m)^k = \begin{pmatrix} 0 & 0 \\ 0 & D_\alpha^k \end{pmatrix}.$$

Thus

$$\exp(\lambda\theta^m) = I + \sum_{k>0} (\theta^m)^k / k! = \Phi^{m,\lambda}.$$

The vectors θ_m form a basis of the Lie algebra of R_u because they are linearly independent and the number of them equals the dimension of R_u . Therefore, a small open neighborhood of identity in R_u , thus R_u itself, is generated by $\widetilde{\Phi^{m,\lambda}}$ for $m \in R_u(\Delta)$.

Combining these with the fact $\widetilde{\text{Aut}}_0 X = G_s \ltimes R_u$ (Proposition 4.14), we get what we want. $\square$

Remark 4.19. Example 3.18 clearly shows the phenomenon discussed in this subsection:

There are four roots in $R(\Delta)$, $\{(0, 1), (1, 0), (-1, 1), (1, -1)\}$. The latter two roots are semi-simple. Using the notation there, the roots $(0, 1), (1, 0)$ correspond to the divisors $E + D'_i - D_0$ and the pairs (x_0, yx'_i) respectively. The one-parameter groups associated with them are exactly the first two listed in Example 3.18. The latter two roots $(-1, 1), (1, -1)$ correspond to the divisors $D'_1 - D'_2, D'_2 - D'_1$ and the pairs $(x'_1, x'_2), (x'_2, x'_1)$ respectively. They are associated with the latter two one-parameter groups listed in Example 3.18.

4.4 Step 4. The key point in the proof of Proposition 4.4

Finally, we come to the part that was skipped before: understanding the relation between $\widetilde{\text{Aut}} X$, $\widetilde{\text{Aut}}_0 X$ and $\text{Aut} X$. The key point is an observation:

Proposition 4.20. *For any $\phi \in \text{Aut} \widetilde{X}$ ($\widetilde{X} = \mathbb{C}^{\Delta_1} \setminus V(\text{Ir}_\Delta)$), ϕ is in $\widetilde{\text{Aut}}_0 X$ (Definition 4.6) if and only if it induces an automorphism on X , still denoted by ϕ , which preserves all divisor classes in $\text{Cl}(X)$ (Definition 3.1). Formally, there is an exact sequence*

$$1 \longrightarrow G \longrightarrow \widetilde{\text{Aut}}_0 X \longrightarrow \text{Aut} X \longrightarrow \text{Aut Cl}(X). \quad (8)$$

Proof. One side of this is clear: $\widetilde{\text{Aut}}_0 X$ is connected and $\text{Aut Cl}(X)$ is discrete, so the image of $\widetilde{\text{Aut}}_0 X$ in $\text{Aut Cl}(X)$ must be the identity. Now we show that every automorphism ϕ on X that preserves all divisor classes can be lifted to $\widetilde{\text{Aut}}_0 X$.

We denote by ϕ^* the pull-back map of ϕ . Since $\phi^*(D_v)$ has the same degree as D_v , the divisor $\phi^*(D_v) - D_v$ is a principal divisor defined by a rational function u_v on X . Pulling back u_v to $\widetilde{X}$ (see Theorem 3.11) and multiplying by x_v , we get a regular function on $\widetilde{X}$, thus on $\mathbb{C}^{\Delta_1}$. Then we can define

$$\widetilde{\phi}_v := u_v x_v$$

and because u_v is G -invariant, $\widetilde{\phi}_v$ is a polynomial of the same degree as x_v .

We now combine $\widetilde{\phi}_v$ to obtain an automorphism $\widetilde{\phi}$. By definition, we have $\widetilde{\phi} \in \widetilde{\text{Aut}}_0 X$, so $\widetilde{\phi}$ descends to an automorphism, also denoted by $\widetilde{\phi}$, on X . However, it does not always coincide with the automorphism ϕ . Luckily, we still have some room in $\widetilde{\phi}_v$ as we can multiply it by a non-zero constant c_v .

We first note that

$$\frac{m}{\widetilde{\phi}^* m} = \prod \frac{x_v^{\langle m, v \rangle}}{\phi^* x_v^{\langle m, v \rangle}} = \prod \frac{1}{u_v} \quad (9)$$

and

$$\left(\frac{m}{\phi^* m}\right) = \sum D_v^{\langle m, v \rangle} - \sum \phi^* D_v^{\langle m, v \rangle} = \left(\prod \frac{1}{u_v}\right),$$

so $\phi^* m$ and $\widetilde{\phi}^* m$ define the same divisor. Hence, $\widetilde{\phi}^* m / \phi^* m$ is a constant. Therefore, $\widetilde{\phi}$ gives an element $t \in T'_X \cong \text{Hom}_{\mathbb{Z}}(M, \mathbb{C})$ by sending $m \in M$ to $\widetilde{\phi}^* m / \phi^* m$.

For an element $c = (c_1, \dots, c_v, \dots) \in (\mathbb{C}^*)^{\Delta(1)}$, we can adjust $\tilde{\phi}$ by defining

$$c\tilde{\phi} := (c_v\tilde{\phi}_v).$$

Recalling the exact sequence (2), we denote the image of c in T'_X by t_c , then (9) tells us that $c\tilde{\phi}$ corresponds to an element $t_c \cdot t \in \text{Hom}_{\mathbb{Z}}(M, \mathbb{C})$, where $\cdot$ means the multiplication in the group T'_X . By (2), we can find a c' such that $t_{c'} = t^{-1}$.

Now we denote $c'\tilde{\phi}$ by $\tilde{\phi}'$, and $\tilde{\phi}'$ certainly descends to X . We can see $\tilde{\phi}'^* m = \phi^* m$ for all characters $m \in M$, thus ϕ' is lifted from ϕ . $\square$

Using this exact sequence and the connectedness argument, we can prove that

Corollary 4.21. *The image of $\widetilde{\text{Aut}}_0 X$ (Definition 4.6) is the connected component of $\text{Aut } X$ containing the identity (we denote it by $\text{Aut}_0 X$).*

Proof. Considering the exact sequence (8) we just proved. $\text{Aut}_0 X$ is connected, so it must be mapped to the identity of $\text{Aut Cl}(X)$, thus it is contained in the image of $\widetilde{\text{Aut}}_0 X$. On the other hand, $\widetilde{\text{Aut}}_0 X$ must be sent to $\text{Aut}_0 X$ since it is connected. $\square$

Therefore, $\text{Aut } X / \text{Aut}_0 X$ is naturally embedded in $\text{Aut Cl}(X)$.

4.5 Step 5. Showing the role of $\text{Aut } \Delta$

Clearly, the next task is to determine the image of $\text{Aut } X$ in $\text{Aut Cl}(X)$. Example 3.19 brings to mind some natural automorphisms, namely the automorphisms induced by the automorphisms of the lattice N that preserve the fan Δ .

Definition 4.22. The automorphism $\mathcal{P} \in \text{Aut } \Delta$ (Definition 2.20) permutes the vectors in Δ_1 , thus defining an automorphism $\phi_{\mathcal{P}}$ of $\mathbb{C}^{\Delta_1}$ by permuting the coordinates. All such automorphisms in $\text{Aut } \mathbb{C}^{\Delta_1}$ form a group denoted by

$$\mathcal{P}(\Delta).$$

Lemma 4.23. $\mathcal{P}(\Delta)$ can be identified with a subgroup of $\text{Aut } X$.

Proof. By definition, $\mathcal{P} \mapsto \phi_{\mathcal{P}}$ is an isomorphism from $\text{Aut } \Delta$ to $\mathcal{P}(\Delta)$.

Conjugating an element $g \in G \subseteq (\mathbb{C}^*)^{\Delta_1}$ by $\phi_{\mathcal{P}} \in \mathcal{P}(\Delta)$ works as applying the corresponding permutation to the diagonal entries of g . The result is clearly an element of $(\mathbb{C}^*)^{\Delta_1}$, and is actually contained in G since if D, D' are linearly equivalent under a character $m \in M$, then their images under the permutation are linearly equivalent under $\mathcal{P}^*(m)$, where $\mathcal{P}^*$ means the dual action of $\mathcal{P}$ over M . In other words, there is a commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & M & \longrightarrow & \bigoplus_v \mathbb{Z}D_v & \longrightarrow & \text{Cl}(X) \longrightarrow 0 \\ & & \downarrow \mathcal{P}^* & & \downarrow \phi_{\mathcal{P}} & & \downarrow \\ 0 & \longrightarrow & M & \longrightarrow & \bigoplus_v \mathbb{Z}D_v & \longrightarrow & \text{Cl}(X) \longrightarrow 0 \end{array}$$

The left vertical arrow is the dual action of $\mathcal{P}$ on M , the middle vertical arrow is the permutation corresponding to $\phi_{\mathcal{P}}$, and the right vertical arrow is induced by the five lemma. Applying the functor $\text{Hom}_{\mathbb{Z}}(\cdot, \mathbb{C}^*)$, we get an isomorphism of G , which is exactly the conjugation of $\phi_{\mathcal{P}}$. Therefore, we have $\mathcal{P}(\Delta) \subseteq \widetilde{\text{Aut}} X$ and every $\phi_{\mathcal{P}} \in \mathcal{P}(\Delta)$ descends to an automorphism on X . We also use $\phi_{\mathcal{P}}$ to denote these automorphisms. Since $\mathcal{P}(\Delta) \cap G = \text{id}$, we can view $\mathcal{P}(\Delta)$ as a subgroup of $\text{Aut } X$. $\square$

We now show that

Lemma 4.24. *In the exact sequence (8), the image of $\mathcal{P}(\Delta)$ (Definition 4.22) in $\text{Aut Cl}(X)$ (Definition 3.1) coincides with the image of $\text{Aut } X$.*

Proof. By Proposition 4.14 and the exact sequence (2), the natural torus action $T'_X \cong (\mathbb{C}^*)^{\Delta_1}/G$ is a maximal torus in $\text{Aut}_0 X$. For any $\phi \in \text{Aut } X$, $\phi T'_X \phi^{-1}$ is another maximal torus of $\text{Aut}_0 X$, so we can find a $\phi' \in \text{Aut}_0 X$ to send it back to T'_X by Theorem 2.28. Namely,

$$\phi' \phi T'_X \phi^{-1} (\phi')^{-1} = T'_X. \quad (10)$$

$\phi' \phi$ is in the same connected component as ϕ .

We claim that if we let $\phi' \phi$ act on X , we have $\phi' \phi(T_X) = T_X$ in X (here T_X is the n -dimensional orbit of the T'_X -action in X): The complex torus $\phi' \phi(T_X)$ is preserved by the T'_X -action on X by (10). But $\phi' \phi(T_X) \cap T_X \neq \emptyset$, so

$$T'_X(\phi' \phi(T_X) \cap T_X) = T_X,$$

which shows $\phi' \phi(T_X) \subseteq T_X$. But both T_X and $\phi' \phi(T_X)$ are n -dimensional T'_X -orbits, thus $\phi' \phi(T_X) = T_X$.

Next, we find an element $t \in T'_X$ such that $t\phi' \phi(T_X)$ preserves the identity, so $t\phi' \phi$ restricted to T_X is a group automorphism. Therefore, $t\phi' \phi$ induces a linear automorphism of the lattice $M = \text{Hom}_{\mathbb{Z}}(T_X, \mathbb{C}^*)$, the dual of which gives an automorphism on the lattice N , thus an element $\phi_{\mathcal{P}}$ in $\mathcal{P}(\Delta)$. The automorphism induced by $\phi_{\mathcal{P}}$ is actually $t\phi' \phi$ on X since they map all the characters in the same way. Because $\phi_{\mathcal{P}} = t\phi' \phi$ is in the same connected component of ϕ , they have the same image in $\text{Aut Cl}(X)$. $\square$

By now, we can finally complete the proof of Proposition 4.4.

Proof of Proposition 4.4. Lemma 4.3 has shown the exactness in places G and $\widetilde{\text{Aut}} X$. We only need to show that all automorphisms of X can be lifted to $\widetilde{\text{Aut}} X$.

Proposition 4.20 tells us how to lift the automorphisms in the identity component $\text{Aut}_0 X$ to $\widetilde{\text{Aut}}_0 X$. Combining with Lemma 4.24, we have $\text{Aut } X \cong \text{Aut}_0 X \rtimes \mathcal{P}(\Delta)$, but $\text{Aut}_0 X \rtimes \mathcal{P}(\Delta) \subseteq \widetilde{\text{Aut}} X$, so the descending map from $\widetilde{\text{Aut}} X$ to $\text{Aut } X$ is surjective. This completes the last part of Proposition 4.4. $\square$

Furthermore, the image of $\mathcal{P}(\Delta)$ (Definition 4.22) in $\text{Aut Cl}(X)$ is $\mathcal{P}(\Delta)/(\mathcal{P}(\Delta) \cap \widetilde{\text{Aut}}_0 X)$. We claim:

Proposition 4.25. *$\mathcal{P}(\Delta) \cap \widetilde{\text{Aut}}_0 X$ (Definition 4.6) is isomorphic to the Weyl group W of G_s with respect to the maximal torus $(\mathbb{C}^*)^{\Delta_1}$ (see Definition 2.26). Furthermore, we have $\text{Aut } X / \text{Aut}_0 X \cong \mathcal{P}(\Delta)/W \cong \text{Aut } \Delta / \text{Aut}_0 \Delta$ (see Definition 2.24).*

Proof. The Weyl group of G_s in this case can be identified with a subgroup of G_s consisting of matrices whose diagonal blocks are permutation matrices (see Example 2.27). A permutation in $\mathcal{P}(\Delta) \cap \widetilde{\text{Aut}}_0 X$ corresponds to a permutation matrix that preserves the degree, so it is in the Weyl group W of G_s . On the other hand, any element in W clearly preserves the torus $T \subseteq X$. The same argument as in the proof of Lemma 4.24 shows that it induces a linear automorphism on the lattice N that preserves the fan Δ , thus corresponds to an element in $\mathcal{P}(\Delta)$. Since $W \subseteq G_s \subseteq \widetilde{\text{Aut}}_0 X$, we get $\mathcal{P}(\Delta) \cap \widetilde{\text{Aut}}_0 X = W$. Then by Lemma 4.24, $\text{Aut } X / \text{Aut}_0 X \cong \mathcal{P}(\Delta)/W$.

For a semisimple root $m \in R_s(\Delta)$, the reflection $\mathcal{P}_m$ (Definition 2.24) swaps the vectors v, v' (which has same degree) and keeps other $v_i \in \Delta_1$, hence corresponds to a degree-preserving element $\mathcal{P}(\Delta)$ thus an element in $(\Delta) \cap \widetilde{\text{Aut}}_0 X = W$. On the other hand, since W can be identified with a

subgroup of G_s consisting of matrices whose diagonal blocks are permutation matrices (see Example 2.27), and $\mathcal{P}_m$ clearly gives all the elementary permutation matrices (i.e., the matrices that swaps two coordinates) of each diagonal blocks (every pair of toric divisors of the same degree corresponds to a semisimple root by Lemma 4.16), W can be generated by the elements corresponding to $\text{Aut}_0 \Delta$. Thus, we have $\mathcal{P}(\Delta)/W \cong \text{Aut } \Delta / \text{Aut}_0 \Delta$. $\square$

Finally, we can prove the main theorem:

Proof of Theorem 3.14. Propositions 4.14, 4.18, Corollary 4.21, and the exact sequence (2) tell us part (i) of Theorem 3.14.

Combining Propositions 4.4, 4.14, 4.18 and Corollary 4.21 we get part (ii) of Theorem 3.14.

Since $W \cap G$ is the identity, the image of W in G_s/G is the Weyl group of G_s/G with respect to the maximal torus $T'_X = (\mathbb{C}^*)^n$. Therefore, Lemma 4.24 and Proposition 4.25 are exactly part (iii) of Theorem 3.14. $\square$

5 Some Generalizations of Theorem 3.14

Cox's method can be generalized to compute the subgroup $\text{Aut} X$ that leaves some (toric invariant) locus invariant. The section provides some generalization of Theorem 3.14 in this direction.

In this section, Δ denotes a complete fan (Definition 2.3), $X := X_\Delta$ denotes the toric variety defined by Δ (Definition 2.4), and P denotes the polytope associated to the fan (Definition 2.13).

5.1 Automorphism groups of toric log pairs

Definition 5.1. If X is a toric variety and $\mathcal{D}$ is a toric invariant divisor on X , then $(X, \mathcal{D})$ is called a *toric log pair* and $\mathcal{D}$ is called the *boundary divisor*. Denote by

$$\text{Aut}(X, \mathcal{D}) \subseteq \text{Aut} X$$

the subgroup of $\text{Aut} X$ (Definition 2.18) that preserves the divisor $\mathcal{D}$. When X is proper over $\mathbb{C}$, $\text{Aut} X$ admitting a (complex) Lie group structure, and the connected component of $\text{Aut}(X, \mathcal{D}) \subseteq \text{Aut} X$ that contains the identity is denoted by

$$\text{Aut}_0(X, \mathcal{D}).$$

Definition 5.2. Let $(X, \mathcal{D})$ be a toric log pair, where $X = X_\Delta$ is a complete toric variety defined by a fan Δ (Definition 2.4) and $\mathcal{D} = \sum_{v \in \Delta_1} a_v D_v$ (Definition 2.12). Let P be the associated polytope (Definition 2.13) and $R(\Delta)$ be the set of roots (Definition 2.15). The *log roots* of $(X, \mathcal{D})$ are the elements of the set

$$R(X, \mathcal{D}) := \{m \in R(\Delta) : \langle m, v \rangle \neq -1 \text{ if } a_v \neq 0\}.$$

The set $R(X, \mathcal{D})$ consists of two subsets:

$$R_s(X, \mathcal{D}) := -R(X, \mathcal{D}) \cap R(X, \mathcal{D}), \quad R_u(X, \mathcal{D}) := R(X, \mathcal{D}) \setminus R_s(X, \mathcal{D}).$$

The elements in $R_s(X, \mathcal{D})$ are called the *semisimple roots* of $(X, \mathcal{D})$.

Remark 5.3. From Remark 2.16, we can see that $R(X, \mathcal{D})$ consists of the roots of Δ that are not on the faces of P that are vertical to one of the vectors $\mathbb{R}v$ that appear in the sum $\mathcal{D} = \sum_{v \in \Delta_1} a_v D_v$. See Figure 4.

Definition 5.4. For a toric log pair $(X_\Delta, \mathcal{D} = \sum_{v \in \Delta_1} a_v D_v)$ and every $\alpha \in \mathcal{C}$ (Definition 3.5), we set

$$\Delta'_\alpha := \{x_v \in \Delta_\alpha : a_v = 0\}.$$

Definition 5.5. Denote by

$$\text{Aut}(\Delta, \mathcal{D})$$

the subgroup of $\text{Aut}\Delta$ (Definition 2.20) that leaves $\mathbb{R}v$ invariant if $a_v \neq 0$.

With these notions, we now prove Theorem 1.2.

Proof of Theorem 1.2. By Proposition 4.4, we have an exact sequence

$$1 \longrightarrow G \longrightarrow \widetilde{\text{Aut}}(X, \mathcal{D}) \longrightarrow \text{Aut}(X, \mathcal{D}) \longrightarrow 1 \quad (11)$$

where $\widetilde{\text{Aut}}(X, \mathcal{D})$ denotes the preimage of $\text{Aut}(X, \mathcal{D})$ in $\widetilde{\text{Aut}}X$. In particular, the identity component of $\widetilde{\text{Aut}}(X, \mathcal{D})$ (denoted by $\widetilde{\text{Aut}}_0(X, \mathcal{D})$) maps to $\text{Aut}_0(X, \mathcal{D})$ and G must be a maximal torus in $\widetilde{\text{Aut}}_0(X, \mathcal{D})$. Together with the exact sequence (2), we have $T'_X \subseteq \text{Aut}_0(X, \mathcal{D})$.

Since D_v corresponds to the coordinate plane defined by $x_v = 0$, $\widetilde{\text{Aut}}_0(X, \mathcal{D})$ must preserve the coordinate plane $x_v = 0$ if $a_v \neq 0$. Hence, for $\phi \in \widetilde{\text{Aut}}_0(X, \mathcal{D})$, the component ϕ_v is a polynomial that can be divided by x_v .

Now consider the image of $\widetilde{\text{Aut}}_0(X, \mathcal{D})$ under the representation $\phi \mapsto \Phi^\phi$ defined in Definition 4.8. If $v \in \alpha$ (recall that α is a divisor class in $\text{Cl}(X)$) and $a_v \neq 0$, then the entry $\varphi_{D_v, D} = 0$ if x^D (see (4)) cannot be divided by x_v . But x^D has the same degree as x_v , so x_v divides x^D implies $x^D = x_v$. Therefore, we can write the diagonal block Φ_α^ϕ as

$$\Phi_\alpha^\phi = \begin{pmatrix} A'_\alpha & A''_\alpha & B'_\alpha \\ 0 & A'''_\alpha & 0 \\ 0 & 0 & C_\alpha \end{pmatrix}, \quad (12)$$

where A'_α , A''_α and B'_α correspond to coefficients of ϕ_v for $v \in \Delta'_\alpha$ (Definition 5.4) and A'''_α is an invertible diagonal matrix of order $|\Delta_\alpha \setminus \Delta'_\alpha|$. The order of A_α is the cardinality $|\Delta'_\alpha|$.

Let G'_s be the subgroup of $\widetilde{\text{Aut}}_0 X$ consisting of elements with diagonal blocks of the form

$$\begin{pmatrix} A'_\alpha & A''_\alpha & 0 \\ 0 & A'''_\alpha & 0 \\ 0 & 0 & C_\alpha \end{pmatrix}$$

and R'_u be the subgroup of $\widetilde{\text{Aut}}_0 X$ consisting of elements with diagonal blocks of the form

$$\begin{pmatrix} I & 0 & B'_\alpha \\ 0 & I & 0 \\ 0 & 0 & C_\alpha \end{pmatrix}.$$

By Lemma 4.12, there is a split exact sequence

$$1 \longrightarrow R'_u \longrightarrow \widetilde{\text{Aut}}_0(X, \mathcal{D}) \longrightarrow G'_s \longrightarrow 1.$$

Furthermore, let G''_s be the subgroup of $\widetilde{\text{Aut}}_0 X$ consisting of elements with diagonal blocks of the form

$$\begin{pmatrix} A'_\alpha & 0 & 0 \\ 0 & A'''_\alpha & 0 \\ 0 & 0 & C_\alpha \end{pmatrix}$$

and R''_u be the subgroup of $\widetilde{\text{Aut}}_0 X$ consisting of elements with diagonal blocks of the form

$$\begin{pmatrix} I & A''_\alpha & 0 \\ 0 & I & 0 \\ 0 & 0 & C_\alpha \end{pmatrix}.$$

By standard linear algebra, there is a split exact sequence

$$1 \longrightarrow R''_u \longrightarrow G'_s \longrightarrow G''_s \longrightarrow 1.$$

Since $G''_s \cong \prod_{\alpha \in \mathcal{C}} \text{GL}(|\Delta'_\alpha|, \mathbb{C}) \times (\mathbb{C}^*)^{|\Delta_\alpha \setminus \Delta'_\alpha|}$, we have

$$\widetilde{\text{Aut}}_0(X, \mathcal{D}) \cong \left(\left(\prod_{\alpha \in \mathcal{C}} \text{GL}(|\Delta'_\alpha|, \mathbb{C}) \times (\mathbb{C}^*)^{|\Delta_\alpha \setminus \Delta'_\alpha|} \right) \ltimes R''_u \right) \ltimes R'_u \quad (13)$$

$$\cong \left(\prod_{\alpha \in \mathcal{C}} (\text{GL}(|\Delta'_\alpha|, \mathbb{C}) \times (\mathbb{C}^*)^{|\Delta_\alpha \setminus \Delta'_\alpha|}) \right) \ltimes (R''_u \ltimes R'_u). \quad (14)$$

(We can change the order of semi-direct products because they are given by split exact sequences.) Thus, the unipotent radical of $\widetilde{\text{Aut}}_0(X, \mathcal{D})$ and $\text{Aut}_0(X, \mathcal{D})$ is $R''_u \ltimes R'_u$. By the proof of Proposition 4.18, $R''_u \ltimes R'_u$ is generated by one-parameter groups $\phi^{m, \lambda}$ (Definition 4.17) associated with the roots $m \in R_u(X, \mathcal{D})$ (Definition 5.2). (Namely, R'_u corresponds to the roots in $R_u(\Delta) \cap R_u(X, \mathcal{D})$ and R''_u corresponds to other roots in $R_u(X, \mathcal{D})$.) Hence, (ii) is shown. Besides, we have shown that $\text{Aut}_0(X, \mathcal{D})$ is a closed subgroup of $\text{Aut}_0 X$, hence a linear algebraic group, which implies (i).

Further assume $a_v \neq a_{v'}$ when $v \neq v'$ and $a_v, a_{v'} \neq 0$. Then all the automorphisms in $\text{Aut}(X, \mathcal{D})$ must leave the divisors D_v invariant if $a_v \neq 0$. Hence the preimage of $\text{Aut}_0(X, \mathcal{D})$ is exactly $\widetilde{\text{Aut}}_0(X, \mathcal{D})$ and the automorphism $\mathcal{P} \in \text{Aut} \Delta \subseteq \text{Aut} X$ (see Lemma 4.23) lies in $\text{Aut}(X, \mathcal{D})$ if and only if $\mathcal{P}$ leaves v invariant, i.e., $\mathcal{P} \in \text{Aut}(\Delta, \mathcal{D})$. Then (iii) follows from Proposition 4.25. $\square$

Corollary 5.6. *For a toric log pair $(X, \mathcal{D})$ where $X = X_\Delta$ is complete, $\text{Aut}(X, \mathcal{D})$ is reductive if and only if $R_u(X, \mathcal{D}) = \emptyset$.*

Some simple examples are listed here. For the pictures of the roots in these example, see Figure 4.

Example 5.7. Denote by L_i the toric invariant lines in $\mathbb{P}^2$. Then $\text{Aut}(\mathbb{P}^2, L_1) \cong \text{GL}(2, \mathbb{C}) \ltimes \mathbb{C}^2$ and $\text{Aut}(\mathbb{P}^2, aL_1 + bL_2) \cong (\mathbb{C}^*)^2 \ltimes \mathbb{C}^2$ when $a \neq b$. They are not reductive.

Denote by L_1, L_2 two parallel toric invariant lines in $\mathbb{P}^1 \times \mathbb{P}^1$. Then $\text{Aut}_0(\mathbb{P}^1 \times \mathbb{P}^1, L_1) \cong (\text{PGL}(2, \mathbb{C}) \times \mathbb{C}^*) \ltimes \mathbb{C}$ is not reductive. But $\text{Aut}_0(\mathbb{P}^1 \times \mathbb{P}^1, aL_1 + bL_2) \cong \text{PGL}(2, \mathbb{C}) \times \mathbb{C}^*$ is reductive.

Let $\text{Bl}_p \mathbb{P}^2$ be $\mathbb{P}^2$ blown up at one toric invariant point p . Denote by L the toric invariant lines in X that does not intersect the exceptional divisor. Then $\text{Aut}(\text{Bl}_p \mathbb{P}^2, L) \cong \text{GL}(2, \mathbb{C})$ is reductive.

5.2 Automorphism groups of certain incomplete toric varieties

Let X be a complete toric variety defined by a fan Δ , and X' be a toric-invariant open subvariety of X such that $\text{codim} X \setminus X' > 1$. Then X' is a toric variety defined by a fan Δ' (not complete) with the same 1-dimensional cones of Δ , and we can compute the automorphism group of X' using a very similar method.

Definition 5.8. For a (possibly incomplete) fan Δ , a lattice point $m \in M$ is the *Demazure root* of Δ if it satisfies the following:

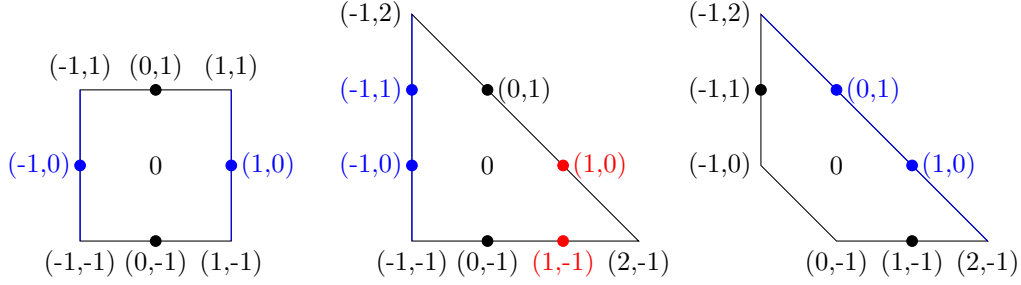


Figure 4: The polytope P corresponding to the toric surfaces $\mathbb{P}^1 \times \mathbb{P}^1$, $\mathbb{P}^2$ and $\mathbb{P}^2$ blown up at 1 point. Take toric log pairs $(\mathbb{P}^1 \times \mathbb{P}^1, aL_1 + bL_2)$, $(\mathbb{P}^2, L_1)$ and $(\text{Bl}_p \mathbb{P}^2, L)$ as in Example 5.7. The points of $R(\Delta)$ are marked along the boundary of P . The points in $R(\Delta) \setminus R(X, \mathcal{D})$ are marked in blue, and the points in $R_u(X, \mathcal{D})$ are marked in red. The edges that correspond to the boundary divisor $\mathcal{D}$ are drawn in blue.

- there exists a unique $v \in \Delta_1$ (Definition 2.12) such that $\langle m, v \rangle = -1$,
- $\langle m, v' \rangle > -1$ for any $v' \in \Delta_1, v' \neq v$,
- for any $\sigma \in \Delta$ lying in $m^\perp$, the cone spanned by σ and $\mathbb{R}_+ v$ is also contained in Δ .

By slightly abusing of the notation, the set of all Demazure roots of Δ is also denoted by

$$R(\Delta).$$

The set $R(\Delta)$ consists of two subsets:

$$R_s(\Delta) := -R(\Delta) \cap R(\Delta), \quad R_u(\Delta) := R(\Delta) \setminus R_s(\Delta).$$

The elements in $R_s(\Delta)$ are called the *semisimple roots* of Δ .

Noting that the third condition in Definition 5.8 is automatically satisfied when Δ is complete, this definition is compatible with Definition 2.15.

Now we provide a sketch of the proof of Theorem 1.3.

Sketch of proof of Theorem 1.3. Since X is normal and $\text{codim} X \setminus X' \geq 2$, by the Hartogs extension theorem [30, 13.5.27, p. 385], any automorphism $f : X' \rightarrow X'$ can extend to a global endomorphism $\bar{f} : X \rightarrow X$ uniquely. And

$$\bar{f} \circ \bar{f}^{-1} = \bar{f} \circ \bar{f}^{-1} = \text{id}_X$$

by the uniqueness, which shows that $\bar{f} \in \text{Aut } X$. Thus, $\text{Aut } X'$ is the subgroup of $\text{Aut } X$ that keeps $X \setminus X'$ invariant, hence a closed subgroup. Therefore, $\text{Aut } X'$ is a linear algebraic group.

By Proposition 4.4, we have an exact sequence

$$1 \longrightarrow G \longrightarrow \widetilde{\text{Aut}} X' \longrightarrow \text{Aut } X' \longrightarrow 1$$

where $\widetilde{\text{Aut}} X'$ denotes the preimage of $\text{Aut } X'$ in $\widetilde{\text{Aut}} X$. In particular, the identity component of $\widetilde{\text{Aut}} X'$ (denoted by $\widetilde{\text{Aut}}_0 X'$) maps to $\text{Aut}_0 X'$. Clearly, the maximal torus T_X^n is contained in $\text{Aut } X'$.

Denoting the preimage of X' in $\tilde{X}$ by $\tilde{X}'$, $\widetilde{\text{Aut}}_0 X'$ is the subgroup of $\widetilde{\text{Aut}}_0 X$ that preserves all irreducible components of $\tilde{X} \setminus \tilde{X}'$. By Proposition 4.18, we should check every one-parameter subgroup $\phi^{m, \lambda}$ associated with $m \in R(\Delta)$ (Definition 4.17). Suppose $(m) = D - D_{v_0}$ and $\sigma \in \Delta \setminus \Delta'$,

let $\tilde{X}_\sigma$ denote the preimage of the closure of the toric orbit corresponding to σ in $\tilde{X}$, then the condition that $\phi^{m,\lambda}$ keeps the $\tilde{X}_\sigma$ invariant means either (i) $\tilde{X}_\sigma$ is not contained in the coordinate plane ($x_{v_0} = 0$); or (ii) $\mathbb{R}_+ v_0$ is a face of σ and x^D can be divided by one of those variables x_v such that $\tilde{X}_\sigma \subseteq (x_v = 0)$. The first case means that $\mathbb{R}_+ v_0$ is not a face of σ ; in the second case, $\sigma_1 \setminus v_0$ cannot span a cone lying in $m^\perp$. Noting that these are exactly the third condition in the definition of $R(\Delta')$ (Definition 5.8), we have $\phi^{m,\lambda} \in \widetilde{\text{Aut}}_0 X'$ if and only if $m \in R(\Delta')$. Then the proposition follows from standard arguments in the Lie group and Lie algebra theory. (Another way is to run an explicit matrix group argument similar to the proof of Proposition 4.14 and Theorem 1.2.) $\square$

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